Combustion and Flame 161 (2014) 84–100

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Experimental and kinetic modeling study on methylcyclohexane pyrolysis and combustion

Zhandong Wang [a], Lili Ye [a], Wenhao Yuan [b], Lidong Zhang [a], Yizun Wang [b], Zhanjun Cheng [a], Feng Zhang [a,*], Fei Qi [a,b,*]

[a] National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, Anhui 230029, PR China
[b] State Key Laboratory of Fire Science, University of Science and Technology of China, Hefei, Anhui 230026, PR China

ARTICLE INFO

Article history:
Received 1 May 2013
Received in revised form 14 August 2013
Accepted 14 August 2013
Available online 12 September 2013

Keywords:
Methylcyclohexane
Pyrolysis
Premixed flame
Kinetic modeling
Synchrotron VUV photoionization mass spectrometry

ABSTRACT

Methylcyclohexane is the simplest alkylated cyclohexane, and has been broadly used as the representative cycloalkane component in fuel surrogates. Understanding its combustion chemistry is crucial for developing kinetic models of larger cycloalkanes and practical fuels. In this work, the synchrotron vacuum ultraviolet photoionization mass spectrometry combined with molecular-beam sampling was used to investigate the species formed during the pyrolysis of methylcyclohexane and in premixed flame of methylcyclohexane. A number of pyrolysis and flame intermediates were identified and quantified, especially including radicals (e.g. CH_3, C_3H_3, C_3H_5 and C_5H_5) and cyclic C6- and C7-intermediates (benzene, 1,3-cyclohexadiene, cyclohexene, toluene, C_7H_{10} and C_7H_{12}, etc.). In particular, the observation of cyclic C6- and C7-intermediates provides important experimental evidence to clarify the special formation channels of toluene and benzene which were observed with high concentrations in both pyrolysis and flame of methylcyclohexane. Furthermore, the rate constants of H-abstraction of methylcyclohexane via H attack, and the isomerization and decomposition of the formed cyclic C_7H_{13} radicals were calculated in this work. A kinetic model of methylcyclohexane combustion with 249 species and 1570 reactions was developed including a new sub-mechanism of MCH. The rate of production and sensitivity analysis were carried out to elucidate methylcyclohexane consumption, and toluene and benzene formation under various pyrolytic and flame conditions. Furthermore, the present kinetic model was also validated by experimental data from literatures on speciation in premixed flames, ignition delays and laminar flame speeds.

1. Introduction

Detailed kinetic modeling of practical fuels is helpful to increase the fuel efficiency and reduce harmful emissions in engines. However, such modeling is challenging since practical fuels are complex mixtures of hundreds of compounds. Instead, surrogate fuels with well-defined and reproducible composition including alkanes, cycloalkanes, and aromatic compounds, etc. were proposed to substitute practical fuels for the development of kinetic models.

Recently, cycloalkanes have received increasing attention because of their high concentrations in practical fuels, for example, they make up 20–40% of typical North American diesel fuel [1,2]. The cycloalkanes with C6 rings tend to dehydrogenate to form aromatics and consequently may increase soot formation [3–10]. Among them, methylcyclohexane (MCH) is the simplest alkylated cyclohexane, and it has been selected as the representative cycloalkane compound in several proposed jet-fuel surrogates [11–17]. MCH was also proposed as an endothermic fuel to cool supersonic aircraft via dehydrogenation to form toluene and hydrogen under the presence of catalyst [18].

Previous studies on MCH include the kinetics of its thermal decomposition using the technique of very low pressure pyrolysis (VLPP) [19], pyrolysis and oxidation of MCH and MCH/toluene blends in a turbulent flow reactor [20], a non-premixed diffusion methane/air flame doped with MCH [21], low- and intermediate-temperature oxidation in motored engines [22], ignition delay time measurement in rapid compression machine (RCM) at various pressures [23,24], and in shock tube at different temperatures and pressures [25–29]. Also, OH concentration time-histories were measured behind reflected shock waves in a shock tube [30] and laminar flame speeds were obtained at the range from 1 to 10 atm [31,32]. Rate constants of the H-abstraction from MCH via OH radical were measured and calculated by Sivaramakrishnan and Michael at high temperature [33]. A recent work by Skeen et al. [8] studied the low pressure laminar premixed flames of MCH at various equivalence ratios exploring the chemistry of MCH

* Corresponding authors. Address: National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, Anhui 230029, PR China. Fax: +86 551 65141078.
E-mail addresses: feng2011@ustc.edu.cn (F. Zhang), fqi@ustc.edu.cn (F. Qi).

0010-2180/$ - see front matter © 2013 The Combustion Institute. Published by Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.combustflame.2013.08.011

decomposition and the formation of aromatic species (e.g. benzene and toluene). More recently, Bounaceur et al. [34] developed a kinetic model to describe the thermal cracking of MCH, which focused on identification of the main consumption and production pathways, in particular the aromatization pathways. Although some aspects of MCH combustion have been studied previously, more reliable data on its pyrolysis and flame are desired including detailed quantitative species concentrations.

In this work, MCH pyrolysis at various pressures (30, 150 and 760 Torr), and premixed flame with equivalence ratio of 1.75 at 30 Torr were investigated using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). The pyrolysis and flame species were identified and their mole fractions were measured. A kinetic model with 249 species and 1570 reactions was developed based on our cyclohexane pyrolysis model [9] by adding a new sub-mechanism of MCH. The model was validated by the experimental results in this work, and premixed flames [8], the ignition delay times [26] and laminar flame speeds [32] reported in the literature. The reaction flux of MCH in pyrolytic and flame conditions was discussed with special attention to the formation of toluene and benzene based on the kinetic simulation.

2. Experimental method

The experimental work was performed at National Synchrotron Radiation Laboratory (NSRL), Hefei, China. The beamline has been described in detail elsewhere [35]. Only brief description of the pyrolysis and flame apparatus is given herein [9,36–38]. The pyrolysis apparatus included a high temperature furnace mounted in the pyrolysis chamber. The pressure of the pyrolysis chamber was controlled by a MKS throat valve at pressures of 30, 150 and 760 Torr. The MCH sample was purchased from the Aladdin Reagent (Shanghai) Co., Ltd with a purity of 99%, which was vaporized in an Ar stream before flowing into the alumina flow reactor (with inner diameter 6.8 mm, and surface to volume ratio $\sim$6.0 cm^{-1}). The total flow rate of the mixture was kept to be 1.0 standard liter per minute (SLM) at 273.15 K under the three pressures, thus the inlet mole fraction of MCH is 0.02. The heating region of the flow reactor is 150 mm. The temperature profiles of the flow reactor under different heating temperatures were measured by a S-type thermocouple with a Ar flow of 1.0 SLM. Figure 1 shows the five temperature profiles measured at 30 Torr with different heating powers. In a test experiment, almost invariable temperature profiles were observed at both low and atmospheric pressures, which can be explained by that the Nusselt number in a laminar flow tube is generally unrelated with pressure, especially in the pressure region of this work. Therefore the temperature profiles are also used to simulate experiments at 150 and 760 Torr. The temperature profile is named by its maximum temperature (T_{max}), which is used as the experimental temperature. The uncertainty of T_{max} is estimated to be within ±30 K, while those at other positions are expected to be less. Since the initial total flow rate is kept constant, the residence times vary with temperature and pressure. The calculated residence times at 30, 150 and 760 Torr in the temperature range studied are 7.4–5.4 $\times$ 10^{-3}, 3.8–2.8 $\times$ 10^{-2} and 2.2–1.6 $\times$ 10^{-1} s, respectively. The method of mole fraction evaluation can be found in the literature [9].

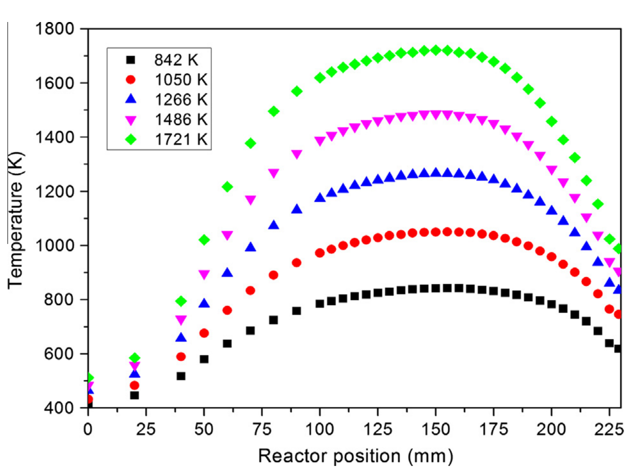


Fig. 1. The temperature profiles along the centerline of the flow tube were measured by moving a S-type thermocouple from the tube inlet to the sampling point of the quartz nozzle. Five temperature profiles with maximum values of 842, 1050, 1266, 1486 and 1721 K measured at 30 Torr are shown.

The premixed flame of MCH with equivalence ratio of 1.75 was stabilized on a McKenna burner with a diameter of 60 mm under 30 Torr. The flow rates of Ar and O_2 were 1.066 and 0.914 SLM, respectively. The flow rate of liquid MCH sample was 0.868 ml/min (equal to 0.152 SLM in the gas phase). The inlet mass flow rate and cold-flow (300 K) velocity were 2.28 $\times$ 10^{-3} g s^{-1} cm^{-2} and 35 cm s^{-1}, respectively. The methods used for species identification and mole fraction evaluation were reported previously [39,40]. The flame temperature profiles were measured by a Pt-6%Rh/Pt-30%Rh thermocouple with a diameter of 0.1 mm coated with Y_2O_3-BeO anti-catalytic ceramic [41,42]. Radiation heat loss [43] and cooling effects of sampling nozzle [44] were calibrated. The uncertainty of the maximum flame temperature is estimated to be ±100 K. We estimate the uncertainty of the flame temperature at other positions is less than that of the maximum flame temperature.

The photoionization cross sections (PICSs) of most species in this work were measured in the literature [45–52] and are available in our online database [53]. For species with unknown values, the PICSs were estimated from molecules with similar structures, for example, 2-methyl-1,3-butadiene refers to 3-methyl-1,2-butadiene [52], *c*C_7H_8 (*o*-isotoluene and *p*-isotoluene) refers to toluene [48], cyclic C_7H_{10} refers to 1,3-cyclohexadiene [47], cyclic C_7H_{12} refers to cyclohexene [47], and 2-heptene refers to 2-hexene [52]. Besides, the PICSs of cyclopentadienyl radical (C_5H_5), cyclopentadiene (C_5H_6) and fulvene were taken from the estimated values in the literature [54,55]. In the pyrolysis experiments, the uncertainties of the measured mole fractions are estimated to be ±10% for H_2 and CH_4, ±25% for the products with known PICSs and a factor of 2 for those with estimated PICSs. The experimental uncertainties of the total C and H balances are around ±10% compared to inlet fluxes of C and H elements. In the premixed flame data, the uncertainties of evaluated mole fractions are about ±10% for major species, ±25% for intermediates with known PICSs, and a factor of 2 for those with estimated PICSs. It should be noted that it is possible that the errors for some species could be even larger.

3. Theoretical calculation

H attack reactions usually play significant roles in flames. These reaction pathways of MCH were computed by the CBS-QB3 method [56] in this work. This composite method starts from a B3LYP/6-311G(2d,d,p) geometry optimization and vibrational analysis, then the single point energy is corrected by a series of high accuracy methods including a complete basis set extrapolation [57]. The quantum chemistry calculations were performed using the Gaussian 09 program [58]. The calculated reaction pathways are shown in Fig. s1 in the supplementary material. Based on these reaction pathways, the temperature-dependent rate constants were computed by the transition state theory (TST). The vibrational mode corresponding to the internal rotation of the methyl group was treated as a hindered rotor with a symmetric hindrance potential function. The rate constants were computed using the ChemRate program [59], and are listed in Table 1. The calculated rate constants of the H-abstraction of MCH by H attack are shown in Fig. 2a–d, while the data used in the work of JetSurF 2.0 [60] and Orme et al. [26] were also shown for comparison. It can be seen that there is no big difference among the three sets of rate constants; however our calculated results are much close to those in JetSurF 2.0.

The H-abstraction of MCH forms five cyclic C7H13 radicals. Further isomerization and decomposition reactions of these radicals were also calculated at CBS-QB3. The intrinsic reaction coordinate (IRC) [56] method was used to confirm that the transition state was correctly connected with its corresponding reactant and products. The calculated reaction pathways are shown in supplementary. Reactions with distinct barriers were treated with the conventional TST, while the variational TST was applied for the barrierless reactions [61]. For the latter, during direct bond breaking, the critical structure which controls the kinetics varies with temperature. The canonical variational high-pressure-limit (HPL) rate constant was computed as a function of both temperature and position. In our calculations, Ar was used as the bath gas. The collisional energy transfer was treated using an exponential-down model with $\langle \text{Edown} \rangle = 0.4\,T$, which is according to our previous work on MCH theoretical calculation [62]. The quantum tunneling effect was neglected in this work considering the comparatively low barriers of reactions involving H atoms (e.g. the H-attack) and the high temperature investigated. Based on the computed reaction pathways, the rate constants were calculated by the RRKM/Master Equation method at 30, 150, 760, 7600, 76,000 Torr in the temperature range of 800–2000 K using the ChemRate program [59], as shown in Table 1.

Figure 2e and f show the rate constants of 4-methyl-cyclohexyl radical isomerization and decomposition. The HPL rate constants of cyclohexyl radical isomerization and decomposition calculated by Sirjean et al. [63] are also shown. The absolute rate constant of each of the two reactions in our calculation is much close to that in Sirjean et al. [63], however the branching ratio of the two reactions is different. In the work of Sirjean et al. [63], the calculated barriers for ring opening and H-loss are 29.5 and 35.3 kcal/mol, respectively. In our calculations for 4-methyl-cyclohexyl radical, the barrier for ring opening is 30.2 kcal/mol and that for H-loss is 33.6 kcal/mol. The difference of barrier heights for the two channels is 5.8 kcal/mol in Sirjean et al. work, while that is 3.4 kcal/mol in this work. This is a possible explanation for the different branching ratio. There are some reports [63–65] on cyclohexyl radical isomerization and decomposition, which is shown in supplementary. Our calculated branching ratio is much close to that in Iwan et al. work [65].

4. Kinetic model

Previously, Orme et al. [26] and Pitz et al. [23] developed the high- and low-temperature oxidation mechanisms of MCH, respectively, which were validated by the ignition delays in shock tube and RCM, and species concentration profiles in a flow reactor. More recently, Wang et al. [60] developed the high-temperature pyrolysis and oxidation mechanism of a series of mono-alkylated cyclohexanes including MCH, which was well validated by laminar flame speeds and ignition delay times in shock tubes, etc. A new kinetic model of MCH was developed in the present work to simulate the experimental data measured in this work (30–760 Torr), and also data reported in the literature (760–7600 Torr) [26,32].

The present kinetic model includes 249 species and 1570 reactions, which was developed from the low pressure pyrolysis model of cyclohexane [9]. For the sub-mechanism of MCH, the rate constants of its H-abstraction by H atom were taken from our calculated results, while H-abstraction by OH, O, O_2, HO_2 and CH_3 adopted the rates from JetSurF 2.0 [60]. The reactions of the formed cyclic C_7H_{13} radicals were calculated in this work. Further decomposition and isomerization of the alkenyl C_7H_{13} radicals were taken from JetSurF 2.0 [60] in the format of Troe parameters. The pressure-dependent rate constants of the initial decomposition and isomerization of MCH were taken from the recent calculation results by Zhang et al. [62], as shown in Table 1. The decomposition rate constants of the formed C_7H_{14} alkenes via allylic C–C bond were referred to the allylic C–C bond dissociation of 1-hexene [66]. To simulate the experimental data at various pressures, the rate constants of the unimolecular reactions were changed to the PLOG format, such as those for the unimolecular reactions of C_3H_5 (aC_3H_5, CH_3CCH_2, CH_3CHCH) [67], C4 species (C_4H_3, C_4H_4, C_4H_5, C_4H_6, C_4H_7, etc.) [66,68], C_5H_5 (cyclopentadienyl radical) [69], cC_6H_{12} (cyclohexane) [66] and C_6H_{12} (1-hexene) [66]. The rate constants of the ring opening and H-loss of cC_6H_{11} (cyclohexyl radical) were referred to 4-methyl-cyclohexyl radical, as shown in Table 1. The rules for H-abstraction from different C positions proposed by Pitz et al. [23] were used to estimate the rate constants of the H-abstraction reactions of C_5H_{10} (1-pentene, 2-pentene), C_6H_{12}, C_6H_{10} (cyclohexene, 1,3-hexadiene, 1,5-hexadiene) and cC_6H_8-13 (1,3-cyclohexadiene), etc. species.

The experimental measurement of a series of cyclic C6 and C7 compounds provides evidence for the specific formation channels of benzene and toluene from MCH. However, these reactions are not included in JetSurF 2.0 [60]. In this work, detailed reaction pathways from MCH to toluene and benzene were developed by including all the important isomers of cyclic C_7H_{12}, C_7H_{11}, C_7H_{10} and C_7H_9 formed in the stepwise dehydrogenation process. The rate constants of the H-elimination of these cyclic compounds were referred to small hydrocarbons, such as C_3H_6 (propene) and $SAXC_4H_7$ (but-1-en-3-yl radical). The rate constants of the H-abstraction of cyclic C_7H_{12} and C_7H_{10} via radical attacks were also estimated following the rate rule of this type of reactions [23]. Besides toluene, the stepwise dehydrogenation of the cyclic C_7H_{13} radicals also forms cC_7H_8 (*o*-isotoluene and *p*-isotoluene). The H-elimination of these two species produces benzyl radical (C_7H_7), their rate constants were adopted from the calculated results by Klippenstein et al. [70].

The thermodynamic data were taken from JetSurF 2.0 [60] and the compilation of Goos, Burcat and Ruscic [71]. The nomenclatures of most species are consistent with JetSurF 2.0 [60] for convenience. There is seldom literature report about the thermodynamic data of the dehydrogenation intermediates from MCH. Thus, their frequencies and standard entropies were computed at CBS-QB3. The internal degrees of freedom except the internal rotations were treated as rigid rotor harmonic oscillators, while hindered rotor treatment was applied for the internal rotations. Their standard enthalpies of formation were calculated by isodesmic reactions while the enthalpies of reactions were calculated at CBS-QB3. Details about the used isodesmic reactions can be found in the Supplementary material. The standard enthalpies of formation of cyclic C_7H_{13} and C_7H_{12} used in Orme et al. model [26] are also shown in Table 2 for a comparison. The discrepancies are about 0.5 kcal/mol for most species. The biggest discrepancy happened in 3-methyl-cyclohexene, which is about 1.3 kcal/mol. Then, the thermodynamic data were fitted to NASA polynomials by the FitDat function of the Chemkin-PRO software [72]. The formulas, nomenclatures, structures and thermodynamic data of these dehydrogenation intermediates are also shown in Table 2.

The simulation was performed with the Plug Flow code for pyrolysis experiment, and the PREMIX code for flame using Chemkin-PRO software [72]. The temperature profiles used for pyrolysis simulation are given in the Supplementary material. The present

Table 1
Sub-mechanism of methylcyclohexane pyrolysis and combustion.

No.	Reactions	*A*	*n*	*E*	Pressure (Torr)	Refs.
Reactions of methylcyclohexane						
1	$CH_3cC_6H_{11} + H = PXCH_2cC_6H_{11} + H_2$	6.66E+06	2.13	8895		This work
2	$CH_3cC_6H_{11} + H = CH_3TXcC_6H_{10} + H_2$	4.53E+06	2.11	4099		This work
3	$CH_3cC_6H_{11} + H = CH_3S2XcC_6H_{10} + H_2$	2.08E+07	2.10	6474		This work
4	$CH_3cC_6H_{11} + H = CH_3S3XcC_6H_{10} + H_2$	2.07E+07	2.12	6441		This work
5	$CH_3cC_6H_{11} + H = CH_3S4XcC_6H_{10} + H_2$	1.16E+07	2.11	6339		This work
6	$CH_3cC_6H_{11} + OH = PXCH_2cC_6H_{11} + H_2O$	5.33E+03	2.90	504		[60]
7	$CH_3cC_6H_{11} + OH = CH_3TXcC_6H_{10} + H_2O$	6.69E+03	2.82	2966		[60]
8	$CH_3cC_6H_{11} + OH = CH_3S2XcC_6H_{10} + H_2O$	8.13E+03	2.84	−1479		[60]
9	$CH_3cC_6H_{11} + OH = CH_3S3XcC_6H_{10} + H_2O$	9.50E+03	2.85	−1023		[60]
10	$CH_3cC_6H_{11} + OH = CH_3S4XcC_6H_{10} + H_2O$	5.82E+03	2.85	−1011		[60]
11	$CH_3cC_6H_{11} + CH_3 = PXCH_2cC_6H_{11} + CH_4$	3.75E+01	3.27	13,516		[60]
12	$CH_3cC_6H_{11} + CH_3 = CH_3TXcC_6H_{10} + CH_4$	1.91E+01	3.27	9022		[60]
13	$CH_3cC_6H_{11} + CH_3 = CH_3S2XcC_6H_{10} + CH_4$	6.68E+01	3.21	11,418		[60]
14	$CH_3cC_6H_{11} + CH_3 = CH_3S3XcC_6H_{10} + CH_4$	7.18E+01	3.26	11,303		[60]
15	$CH_3cC_6H_{11} + CH_3 = CH_3S4XcC_6H_{10} + CH_4$	1.82E+01	3.36	10,931		[60]
16	$CH_3cC_6H_{11} = cC_6H_{11} + CH_3$	1.85E+105	−26.15	129,214	7.6	[62]
		1.19E+92	−22.23	122,595	30	[62]
		1.50E+80	−18.67	116,938	150	[62]
		5.93E+64	−14.15	108,486	760	[62]
		3.59E+45	−8.56	97,430	7600	[62]
		1.66E+33	−4.99	90,100	76,000	[62]
		1.10E+27	−3.20	86,385	HPL	[62]
17	$CH_3cC_6H_{11} = C_7H_{14}$-1	5.49E+115	−29.26	143,180	7.6	[62]
		1.03E+104	−25.68	138,280	30	[62]
		1.92E+88	−20.96	130,954	150	[62]
		5.57E+69	−15.49	121,178	760	[62]
		4.90E+44	−8.17	107,073	7600	[62]
		7.98E+26	−3.02	96,627	76,000	[62]
		1.09E+17	−0.16	90,732	HPL	[62]
18	$CH_3cC_6H_{11} = C_7H_{14}$-2	9.52E+114	−28.98	140,737	7.6	[62]
		1.74E+103	−25.40	135,829	30	[62]
		3.34E+87	−20.68	128,511	150	[62]
		9.66E+68	−15.21	118,735	760	[62]
		8.51E+43	−7.89	104,630	7600	[62]
		1.38E+26	−2.74	94,184	76,000	[62]
		1.90E+16	0.12	88,289	HPL	[62]
19	$CH_3cC_6H_{11} = CH_3$-2-1C_6H_{11}	2.25E+132	−34.10	155,250	7.6	[62]
		1.46E+109	−27.20	144,350	30	[62]
		4.09E+93	−22.49	137,505	150	[62]
		3.27E+74	−16.83	127,710	760	[62]
		3.09E+47	−8.92	112,705	7600	[62]
		2.18E+27	−3.08	100,948	76,000	[62]
		6.04E+15	0.27	94,051	HPL	[62]
20	$CH_3cC_6H_{11} = CH_3$-5-1C_6H_{11}	1.99E+132	−34.11	154,207	7.6	[62]
		1.28E+109	−27.20	143,306	30	[62]
		3.61E+93	−22.49	136,462	150	[62]
		2.89E+74	−16.83	126,667	760	[62]
		2.73E+47	−8.93	111,662	7600	[62]
		1.93E+27	−3.08	99,905	76,000	[62]
		5.34E+15	0.27	93,008	HPL	[62]
21	$CH_3cC_6H_{11} = CH_3$-3-1C_6H_{11}	1.70E+119	−30.34	147,025	7.6	[62]
		1.77E+108	−26.94	142,902	30	[62]
		4.67E+92	−22.23	136,004	150	[62]
		3.71E+73	−16.58	126,164	760	[62]
		5.09E+46	−8.72	111,220	7600	[62]
		6.68E+26	−2.95	99,619	76,000	[62]
		2.53E+15	0.36	92,802	HPL	[62]
22	$CH_3cC_6H_{11} = CH_3$-4-1C_6H_{11}	1.87E+119	−30.34	146,825	7.6	[62]
		1.95E+108	−26.95	142,703	30	[62]
		5.13E+92	−22.24	135,803	150	[62]
		4.07E+73	−16.58	125,964	760	[62]
		5.59E+46	−8.72	111,019	7600	[62]
		7.33E+26	−2.95	99,418	76,000	[62]
		2.78E+15	0.35	92,602	HPL	[62]
Reactions of cC_7H_{13} radicals						
23	$PXCH_2cC_6H_{11} = PXC_7H_{13}$	8.92E+18	−2.94	19,897	30	This work
		6.83E+22	−3.82	23,511	150	This work
		1.56E+29	−5.39	29,211	760	This work
		4.15E+37	−7.44	37,753	7600	This work
		1.92E+39	−7.58	42,286	76,000	This work
		2.53E+13	0.16	29,785	HPL	This work
24	$PXCH_2cC_6H_{11} = CH_2cC_6H_{10} + H$	8.14E+14	−2.42	21,719	30	This work
		1.23E+20	−3.61	26,095	150	This work

(*continued on next page*)

Table 1 (*continued*)

No.	Reactions	A	n	E	Pressure (Torr)	Refs.
		1.18E+28	−5.58	32,946	760	This work
		1.31E+39	−8.28	43,693	7600	This work
		4.67E+42	−8.87	50,298	76,000	This work
		1.60E+11	0.59	35,447	HPL	This work
25	$CH_3TXcC_6H_{10} = CH_3\text{-}2\text{-}PXC_6H_{10}$	7.77E+21	−3.78	22,330	30	This work
		2.46E+23	−3.98	24,528	150	This work
		3.46E+31	−6.06	31,348	760	This work
		3.04E+40	−8.25	40,204	7600	This work
		3.40E+43	−8.78	45,692	76,000	This work
		2.51E+13	0.24	31,344	HPL	This work
26	$CH_3TXcC_6H_{10} = CH_2cC_6H_{10} + H$	5.68E+24	−4.98	28,465	30	This work
		4.63E+25	−4.97	30,017	150	This work
		2.67E+34	−7.20	37,107	760	This work
		4.39E+44	−9.70	46,944	7600	This work
		1.12E+49	−10.55	53,815	76,000	This work
		1.18E+14	−0.04	37,343	HPL	This work
27	$CH_3TXcC_6H_{10} = CH_3\text{-}1\text{-}cC_6H_9 + H$	1.47E+22	−4.09	24,874	30	This work
		1.57E+23	−4.12	26,671	150	This work
		7.83E+31	−6.33	33,844	760	This work
		6.83E+41	−8.76	43,572	7600	This work
		7.93E+45	−9.52	50,174	76,000	This work
		5.80E+12	0.46	34,881	HPL	This work
28	$CH_3S2XcC_6H_{10} = CH_3\text{-}3\text{-}PXC_6H_{10}$	1.24E+22	−4.01	22,066	30	This work
		4.17E+22	−3.93	23,430	150	This work
		1.01E+28	−5.23	28,271	760	This work
		2.14E+37	−7.54	37,062	7600	This work
		1.11E+43	−8.81	44,209	76,000	This work
		1.59E+13	0.22	31,118	HPL	This work
29	$CH_3S2XcC_6H_{10} = PX7\text{–}2C_7H_{13}$	2.99E+22	−4.04	21,843	30	This work
		8.66E+22	−3.95	23,164	150	This work
		1.44E+28	−5.21	27,888	760	This work
		1.61E+37	−7.45	36,452	7600	This work
		4.39E+42	−8.66	43,334	76,000	This work
		2.37E+13	0.19	30,429	HPL	This work
30	$CH_3S2XcC_6H_{10} = CH_3\text{-}1\text{-}cC_6H_9 + H$	2.06E+20	−3.80	23,117	30	This work
		1.50E+21	−3.78	24,651	150	This work
		2.14E+27	−5.27	30,013	760	This work
		8.88E+37	−7.90	39,841	7600	This work
		9.49E+44	−9.50	48,246	76,000	This work
		5.69E+11	0.60	34,004	HPL	This work
31	$CH_3S2XcC_6H_{10} = CH_3\text{-}3\text{-}cC_6H_9 + H$	1.82E+20	−3.68	23,447	30	This work
		1.92E+21	−3.70	25,053	150	This work
		5.62E+27	−5.27	30,610	760	This work
		7.34E+38	−8.02	40,816	7600	This work
		2.67E+46	−9.75	49,698	76,000	This work
		2.01E+12	0.63	35,196	HPL	This work
32	$CH_3S2XcC_6H_{10} = cC_6H_{10} + CH_3$	4.24E+21	−3.89	22,867	30	This work
		2.58E+22	−3.86	24,360	150	This work
		2.27E+28	−5.30	29,571	760	This work
		4.14E+38	−7.84	39,088	7600	This work
		1.94E+45	−9.36	47,105	76,000	This work
		6.35E+13	0.20	33,524	HPL	This work
33	$CH_3S3XcC_6H_{10} = CH_3\text{-}4\text{-}PXC_6H_{10}$	1.21E+18	−2.84	19,903	30	This work
		7.98E+21	−3.70	23,365	150	This work
		2.52E+28	−5.31	28,978	760	This work
		8.05E+37	−7.66	38,033	7600	This work
		6.13E+42	−8.70	44,718	76,000	This work
		1.68E+13	0.21	31,241	HPL	This work
34	$CH_3S3XcC_6H_{10} = SXC_7H_{13}$	5.23E+18	−2.93	19,555	30	This work
		2.06E+22	−3.73	22,876	150	This work
		3.29E+28	−5.27	28,275	760	This work
		3.31E+37	−7.49	36,918	7600	This work
		8.80E+41	−8.42	43,157	76,000	This work
		2.55E+13	0.17	30,061	HPL	This work
35	$CH_3S3XcC_6H_{10} = CH_3\text{-}4\text{-}cC_6H_9 + H$	1.19E+16	−2.44	20,998	30	This work
		5.50E+20	−3.50	24,928	150	This work
		1.93E+28	−5.37	31,247	760	This work
		3.19E+39	−8.14	41,683	7600	This work
		1.03E+46	−9.58	49,952	76,000	This work
		1.87E+12	0.63	34,980	HPL	This work
36	$CH_3S3XcC_6H_{10} = CH_3\text{-}3\text{-}cC_6H_9 + H$	6.78E+14	−2.23	21,396	30	This work
		8.10E+19	−3.39	25,518	150	This work
		8.05E+27	−5.38	32,106	760	This work
		6.73E+39	−8.33	43,061	7600	This work
		1.12E+47	−9.95	51,964	76,000	This work

Table 1 (*continued*)

No.	Reactions	*A*	*n*	*E*	Pressure (Torr)	Refs.
		1.64E+12	0.61	36,658	HPL	This work
37	$CH_3S4XcC_6H_{10} = PXCH_2\text{-}5\text{-}1C_6H_{11}$	1.26E+22	−3.85	22,627	30	This work
		3.57E+26	−4.89	26,638	150	This work
		2.59E+32	−6.32	32,020	760	This work
		2.07E+41	−8.51	40,814	7600	This work
		5.79E+44	−9.15	46,530	76,000	This work
		3.87E+13	0.20	31,782	HPL	This work
38	$CH_3S4XcC_6H_{10} = CH_3\text{-}4\text{-}cC_6H_9 + H$	8.38E+20	−3.63	23,771	30	This work
		8.32E+25	−4.80	28,120	150	This work
		5.73E+32	−6.47	34,206	760	This work
		1.37E+43	−9.02	44,242	7600	This work
		7.21E+47	−9.98	51,272	76,000	This work
		3.78E+12	0.63	34,881	HPL	This work
Reactions of cyclohexyl radical						
39	$cC_6H_{11} = PXC_6H_{11}$	1.26E+22	−3.85	22,627	30	Ref to R37
		3.57E+26	−4.89	26,638	150	
		2.59E+32	−6.32	32,020	760	
		2.07E+41	−8.51	40,814	7600	
		5.79E+44	−9.15	46,530	76,000	
		3.87E+13	0.20	31,782	HPL	
40	$cC_6H_{11} = cC_6H_{10} + H$	8.38E+20	−3.63	23,771	30	Ref to R38
		8.32E+25	−4.80	28,120	150	
		5.73E+32	−6.47	34,206	760	
		1.37E+43	−9.02	44,242	7600	
		7.21E+47	−9.98	51,272	76,000	
		3.78E+12	0.63	34,881	HPL	

Note: $k = AT^n \exp(-E/RT)$. The units are in cm^3, mol, s, cal.
HPL means high pressure limit.

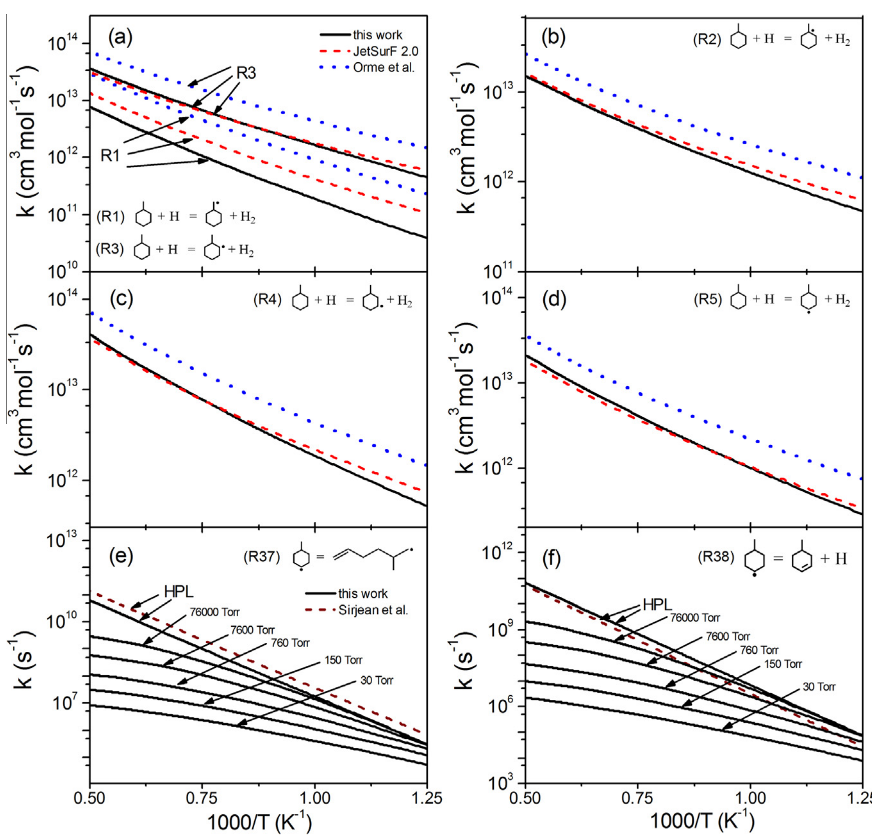
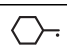
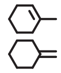
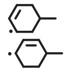
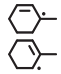
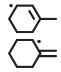
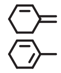
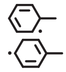
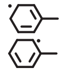
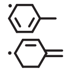


Fig. 2. (a–d) rate constants of the H-abstraction of MCH by H attack calculated in this work, in JetSurF 2.0 [60] and Orme et al. [26]. (e–f) rate constants of the isomerization and decomposition of the 4-methyl-cyclohexyl radical ($CH_3S4XcC_6H_{10}$) calculated in this work. The HPL rate constants of the cyclohexyl radical isomerization and decomposition calculated by Sirjean et al. [63] are shown for a comparison.

Table 2
Thermodynamic properties for the dehydrogenation species from MCH.

Formula	Nomenclature	Structure	$\Delta_f H^0$ (298 K)	$\Delta_f H^0$ (298 K)	S^0 (298 K)	C_p						
						300 K	400 K	500 K	600 K	800 K	1000 K	1500 K
C_7H_{13}	$PXCH_2cC_6H_{11}$		12.75	12.25[a]	84.78	31.69	42.34	52.00	60.15	72.79	81.99	96.02
C_7H_{13}	$CH_3TXcC_6H_{10}$		8.18	7.65[a]	85.59	30.98	41.45	51.13	59.41	72.33	81.73	96.00
C_7H_{13}	$CH_3S2XcC_6H_{10}$		10.17	9.6[a]	84.37	31.48	42.03	51.66	59.85	72.59	81.87	96.00
C_7H_{13}	$CH_3S3XcC_6H_{10}$		9.91	9.6[a]	84.31	31.56	42.07	51.70	59.89	72.61	81.89	96.02
C_7H_{13}	$CH_3S4XcC_6H_{10}$		10.00	9.6[a]	84.21	31.50	42.03	51.66	59.85	72.59	81.89	96.02
C_7H_{12}	CH_3-4-cC_6H_9		−8.67	−9.37[a]	80.61	29.35	39.50	48.76	56.63	68.79	77.64	91.05
C_7H_{12}	CH_3-3-cC_6H_9		−8.19	−9.54[a]	80.66	29.27	39.46	48.72	56.59	68.77	77.62	91.03
C_7H_{12}	CH_3-1-cC_6H_9		−10.12	−10.65[a]	81.23	29.49	39.52	48.70	56.53	68.71	77.58	91.03
C_7H_{12}	$CH_2cC_6H_{10}$		−8.12	−7.71[a]	79.59	28.79	39.25	48.62	56.55	68.75	77.60	91.01
C_7H_{11}	CH_3-5-$SAXcC_6H_8$		22.32		81.84	29.43	39.23	47.93	55.22	66.41	74.46	86.68
C_7H_{11}	CH_3-4-$SAXcC_6H_8$		22.77		81.86	29.35	39.15	47.89	55.18	66.37	74.42	86.64
C_7H_{11}	CH_3-3-$SAXcC_6H_8$		20.65		83.08	29.47	39.07	47.73	55.02	66.25	74.38	86.66
C_7H_{11}	CH_3-2-$SAXcC_6H_8$		22.37		86.10	29.77	39.36	47.97	55.22	66.37	74.42	86.66
C_7H_{11}	CH_3-1-$SAXcC_6H_8$		20.66		83.14	29.47	39.07	47.73	55.02	66.25	74.38	86.66
C_7H_{11}	$CH_2SAXcC_6H_9$		25.40		81.32	29.17	39.13	47.91	55.22	66.39	74.44	86.66
C_7H_{10}	CH_2-3-cC_6H_8		15.96		78.37	27.32	36.78	45.11	52.06	62.63	70.22	81.71
C_7H_{10}	CH_3-2-cC_6H_7-13		16.35		79.94	28.16	37.24	45.39	52.22	62.71	70.30	81.77
C_7H_{10}	CH_3-1-cC_6H_7-13		15.66		79.83	28.08	37.18	45.33	52.18	62.71	70.30	81.77
C_7H_{10}	CH_3-5-cC_6H_7-13		17.99		79.29	27.88	37.16	45.39	52.26	62.79	70.34	81.77
C_7H_9	CH_3-5-$SAXcC_6H_6$		37.45		80.93	27.66	36.44	44.13	50.51	60.17	67.06	77.40
C_7H_9	CH_3-6-$SAXcC_6H_6$		40.38		80.33	27.38	36.40	44.21	50.59	60.21	67.02	77.32
C_7H_9	CH_3-2-$SAXcC_6H_6$		38.68		81.48	27.86	36.64	44.31	50.65	60.25	67.08	77.40
C_7H_9	CH_3-3-$SAXcC_6H_6$		38.85		85.73	27.78	36.50	44.15	50.51	60.17	67.06	77.40
C_7H_9	CH_3-4-$SAXcC_6H_6$		38.68		81.48	27.86	36.64	44.31	50.65	60.25	67.08	77.40
C_7H_9	CH_3-1-$SAXcC_6H_6$		37.42		80.99	27.68	36.46	44.15	50.53	60.19	67.06	77.42
C_7H_9	CH_2-6-$SAXcC_6H_7$		43.07		79.26	27.30	36.42	44.25	50.65	60.25	67.06	77.34
C_7H_9	CH_2-3-$PXcC_6H_7$		50.10		79.79	27.82	36.76	44.51	50.87	60.41	67.16	77.30
C_7H_8	CH_2-5-cC_6H_6-13		42.96		77.81	25.91	34.44	41.73	47.69	56.61	62.95	72.45
C_7H_8	CH_2-3-cC_6H_6-14		39.02		76.68	25.75	34.28	41.57	47.55	56.51	62.89	72.45

Note: the unit for $\Delta_f H^0$, S^0 and C_p is kcal/mol, cal/mol/K and cal/mol/K, respectively.
[a] Refer to the work of Orme et al. [26].

model was also used to re-simulate the low pressure pyrolysis of cyclohexane in our previous work [9]. The comparison between experiment and simulation is shown in supplementary. Similar to previous work, the consumption of cyclohexane and formation of the pyrolysis species are reproduced satisfactorily.

5. Results and discussion

5.1. Pyrolysis of MCH

In the pyrolysis experiment, more than 30 species were identified and quantified, while ~15 species were detected and quantified in the pyrolysis study of MCH by Zippieri et al. [20] with ethylene, 1,3-butadiene, methane and propene as the major products. The new detected and quantified species in this work are the initial isomerization and decomposition products: 2-heptene and methyl radical, cyclic compounds C_7H_{12}, C_7H_{10}, C_7H_8 (*o*-isotoluene and *p*-isotoluene), 1,3-cyclohexadiene and fulvene, etc. 2-Heptene and some cyclic C_7H_{12} compounds were also detected in the stream pyrolysis study of MCH by Pant and Kunzru [73] in a tubular reactor at atmospheric pressure in the temperature range 953–1073 K. However, these experimental observations are quite different to the thermal decomposition of MCH under supercritical conditions by Kim et al. [74].

Figure 3 displays the mole fraction profiles of MCH and its initial decomposition products, e.g. 2-heptene, 1-heptene and CH_3. For the initial products, the methyl radical was only detected in the 30 Torr experiment and was below the limit of detectability at higher pressures. According to the quantum calculations by Zhang et al. [62], 2-heptene is the dominant isomerization product of MCH pyrolysis. In our measurement, 2-heptene was quantified in the 30 Torr experiment, and the simulated result agrees with our measurement. However, 2-heptene was not quantified under 150 and 760 Torr probably due to its low concentration, which is also indicated by the simulation. Other isomerization products like 1-heptene were also not detected due to their very low concentration.

Based on the prediction by the present model, the reaction flux analysis of MCH at 30 and 760 Torr is shown in Fig. 4 with a

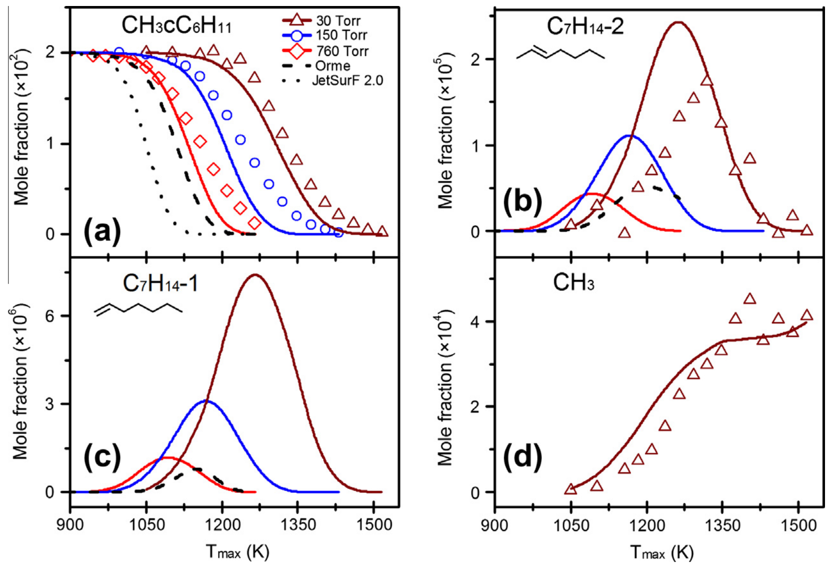


Fig. 3. Measured (symbols) and simulated (lines) mole fraction profiles of (a) MCH, (b) 2-heptene, (c) 1-heptene and (d) CH_3 in MCH pyrolysis at 30, 150 and 760 Torr. Solid lines are the simulated results with the present model, dash and dotted lines are the simulated results by Orme et al. [26] and JetSurF 2.0 [60] models at 760 Torr, respectively.

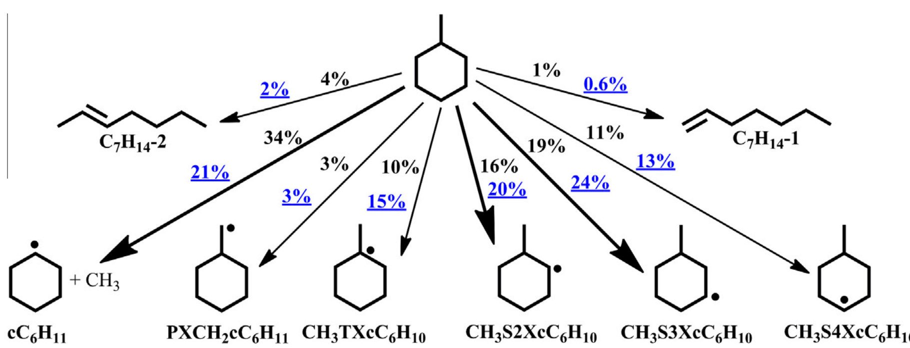


Fig. 4. Reaction flux analysis of MCH consumption in pyrolysis at 30 Torr (in black) and 760 Torr (in blue and underlines). The conversion of MCH at both pressures is approximately 80%. The numbers indicate the carbon flux of the pathway divided by total carbon flux of MCH consumption. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

conversion of approximately 80%. MCH is consumed by decomposition, isomerization and H-abstraction via H and CH_3 attack at both 30 and 760 Torr. At 30 Torr, nearly 40% of MCH is consumed by decomposition and isomerization, of which the decomposition to cyclohexyl and methyl radicals contributes the most. As the pressure increases, H-abstraction by radicals attack forming cyclic C_7H_{13} radicals becomes more important.

Moreover, the sensitivity analysis of MCH consumption at 30 and 760 Torr was performed under the same condition of reaction flux analysis, as shown in Fig. 5. At both pressures, the decomposition and H-abstraction of MCH have positive effect on MCH consumption, among which the CH_3-loss channel has the largest impact on MCH consumption. These reactions have positive effects on MCH consumption, not only because these reactions directly consume MCH, but also because the further decomposition of formed cyclic C_6H_{11} and C_7H_{13} radicals are the source of H atom via the H-loss of C_2H_5 and C_4H_7, etc. The formation of H atom further promotes the H-abstraction reactions. This explains the positive effect of H-loss reactions of C_2H_5 and C_4H_7 on MCH consumption.

Furthermore, the kinetic models developed by Orme et al. [26] and JetSurF 2.0 [60] were also used to simulate the experimental data at 760 Torr. The prediction of MCH at 760 Torr by Orme

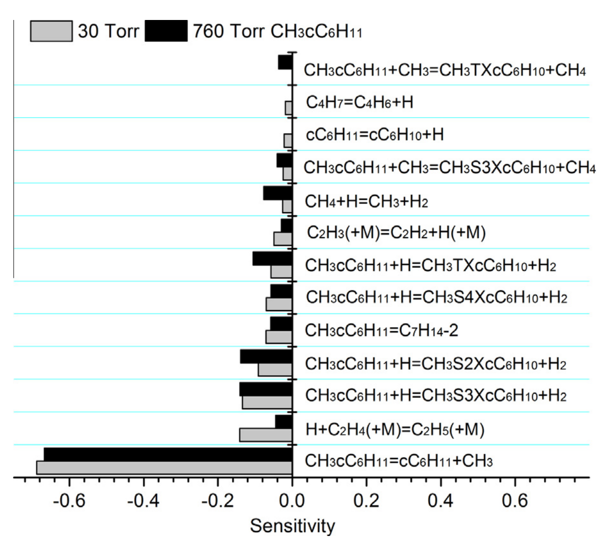


Fig. 5. Normalized sensitivity analysis of MCH consumption in pyrolysis at 30 and 760 Torr. The conversion of MCH is approximately 80%. Reactions with absolute sensitivity coefficients larger than 0.02 and 0.03 are considered for 30 and 760 Torr, respectively.

et al. [26] (shown as dash lines in Fig. 3) is close to our simulated result while the formation of 2-heptene and 1-heptene starts at higher temperature. These differences are probably caused by their different formation channels between the two models. In Orme et al. model [26], the ring-opening reactions of MCH were taken from the work of Brown et al. [19], where three C_7H_{14} biradicals were considered. According to studies by Skeen et al. [8] and Zhang et al. [62], these biradicals undergo internal H-migration forming heptenes and methyl-hexenes, and the direct C–C scissions are energy unfavorable. Both internal H-migration and C–C scission reactions were considered in Orme et al. [26] model. However, these biradicals are mainly consumed by the unfavorable C–C scission channels. For the simulation by JetSurF 2.0 [60] (see the dotted line in Fig. 3), the consumption of MCH is significantly over-predicted, which is probably caused by the too large rate constants of MCH isomerizations.

As shown in Fig. 4, the decomposition and H-abstraction of MCH form cyclohexyl and cyclic C_7H_{13} radicals. They were not detected in this work due to their instability. The ring opening and decomposition of these cyclic compounds produce various intermediates detected in this work. The following discussion is organized according to their reaction pathways.

5.1.1. Ring opening of the cyclohexyl and cyclic C_7H_{13} radicals

The cyclohexyl and cyclic C_7H_{13} radicals are mainly consumed via ring-opening isomerization, as shown in Fig. 6. The ring opening of cyclohexyl radical forms hex-5-en-1-yl radical (PXC_6H_{11}), which undergoes internal 1,4 H-migration forming hex-1-en-3-yl radical ($SAXC_6H_{11}$) and direct β-C–C scission. Besides these two channels, the isomerization of PXC_6H_{11} forms a five-member-ring product $CH_2cC_5H_9$ ( ) [64]. In our kinetic model, the rate constant of this reaction was referred to the inverse reaction of R23 in Table 1. Further reactions of $CH_2cC_5H_9$ undergoes H-loss forming $CH_2cC_5H_8$ ( ) and internal H-migration forming $CH_3cC_5H_8$ ( ) [75]. The rate constant of the H-loss was referred to R24 in Table 1, and that of H-migration was taken from JetSurF 2.0 [60]. Furthermore, the ring-opening reaction of $CH_3cC_5H_8$ was referred to that of 3-methyl-cyclohexyl radical ($CH_3S3XcC_6H_{10}$) in Table 1. Our simulation shows that hex-5-en-1-yl radical (PXC_6H_{11}) is dominantly consumed via H-migration forming hex-1-en-3-yl radical ($SAXC_6H_{11}$) and direct β-C–C scission while the channel to $CH_2cC_5H_9$ is negligible.

Similar to the cyclohexyl radical, the ring opening of cyclic C_7H_{13} radicals produces alkenyl C_7H_{13} radicals. The alkenyl C_7H_{13} radicals are consumed by two kinds of reactions: the internal H-migration forming conjugate intermediates (not shown in Fig. 6), which are then rapidly decomposed by β-C–C scission yielding dienes and small radicals; the direct β-C–C scission producing alkenes and small radicals. Some of these alkenyl C_7H_{13} radicals can also undergo isomerization forming five-member-ring products. This kind of reactions is similar to that of hex-5-en-1-yl radical (PXC_6H_{11}). Considering the negligible contribution of the five-member-ring isomerization channel in hex-5-en-1-yl radical consumption, these reactions for alkenyl C_7H_{13} radicals were not included in the present model. Some quantified C0–C6 species are shown in Fig. 7 with simulated results. The reaction flux for the cyclohexyl and cyclic C_7H_{13} radicals at 760 Torr is similar to that in 30 Torr. Thus, we only discuss the reaction flux at 30 Torr for illustration.

Figure 7a shows the mole fractions of 1,3-hexadiene (C_6H_{10}-13), its maximum concentration at 30 Torr is about three times higher than that in cyclohexane pyrolysis [9]. In cyclohexane pyrolysis, 1,3-hexadiene is dominantly produced from hex-5-en-1-yl radical (PXC_6H_{11}). However, this channel is negligible in MCH pyrolysis.

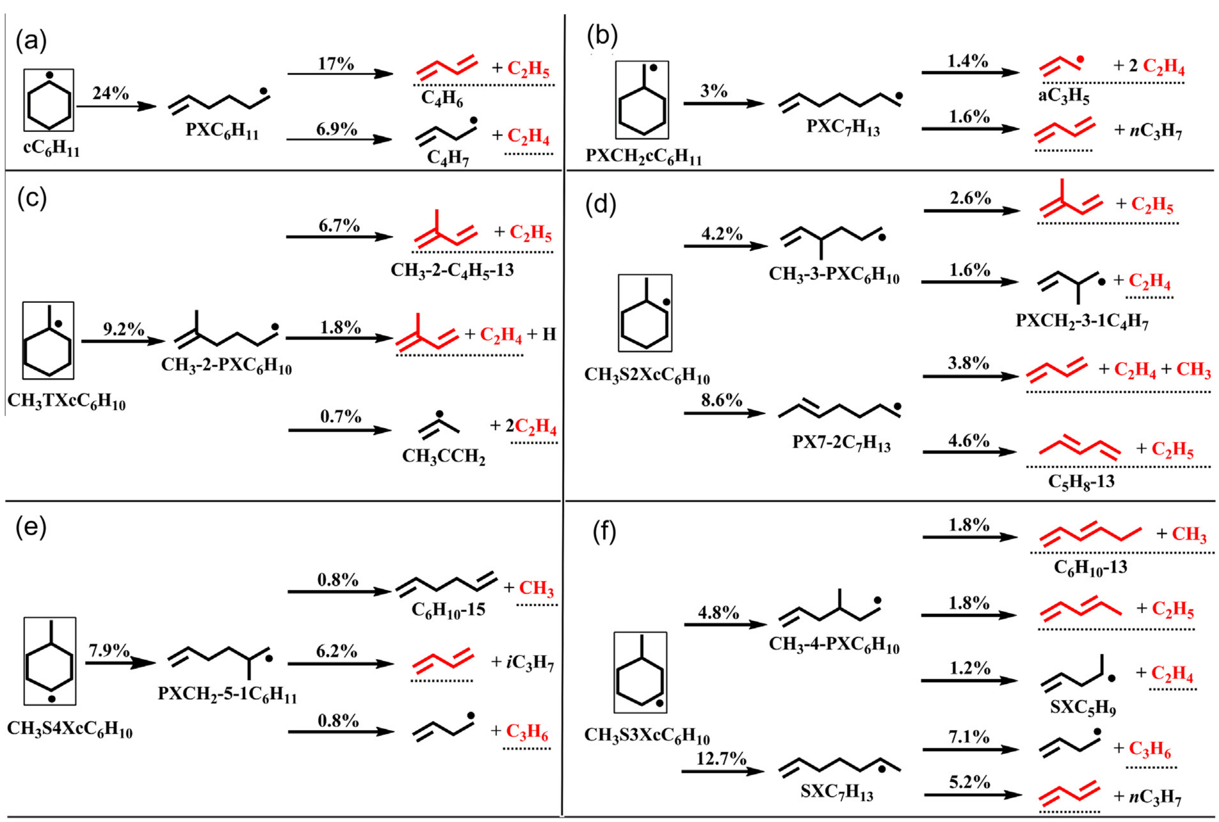


Fig. 6. Reaction flux analysis of the ring opening of cyclohexyl and cyclic C_7H_{13} radicals in pyrolysis at 30 Torr. The conversion of MCH is approximately 80%. The numbers indicate the carbon flux of the pathway divided by total carbon flux of MCH consumption. Species labeled with red and underlines were detected in the experiment.

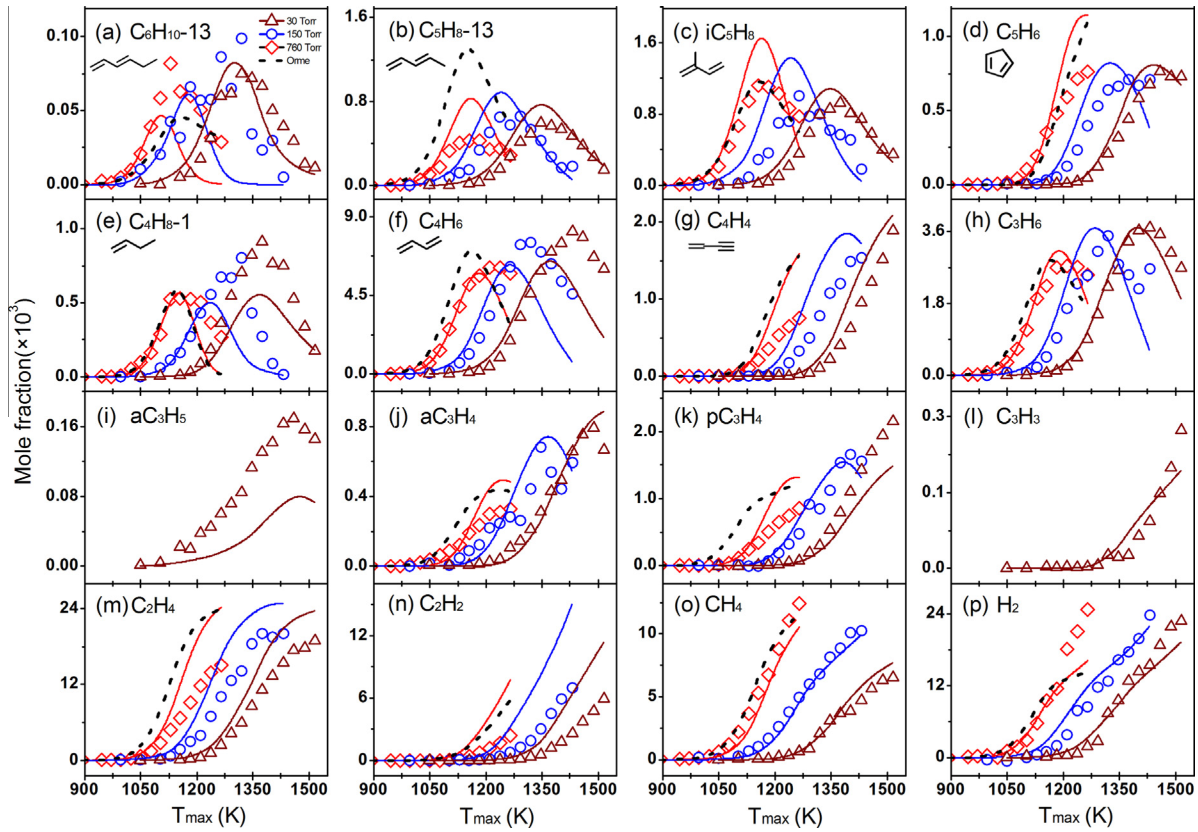


Fig. 7. Measured (symbols) and simulated (lines) mole fraction profiles of C0–C6 products in MCH pyrolysis at 30, 150 and 760 Torr. Solid lines are the simulated results with the present model, while dash lines are the simulated results by Orme et al. [26] model at 760 Torr.

Instead, it is dominantly produced from the decomposition of 3-methyl-hex-5-en-1-yl radical (CH_3-4-PXC_6H_{10}) coming from 3-methyl-cyclohexyl radical (CH_3S3XcC_6H_{10}). Besides 1,3-hexadiene, two C5 dienes, i.e. 1,3-pentadiene (C_5H_8-13) and 2-methyl-1,3-butadiene (CH_3-2-C_4H_5-13) were detected and quantified, as shown in Fig. 7b and c. Their summarized maximum mole fraction is about 3 times higher than that in cyclohexane pyrolysis [9]. In cyclohexane pyrolysis, 2-methyl-1,3-butadiene and 1,3-pentadiene are formed mainly from the build-up reactions from smaller intermediates while these two species come from the decomposition of cyclic C_7H_{13} radicals in MCH pyrolysis. There exist two main pathways for 2-methyl-1,3-butadiene formation: the decomposition of 3-methyl-hex-1-en-6-yl radical (CH_3-3-PXC_6H_{10}, 24%) and 2-methyl-hex-1-en-6-yl radical (CH_3-2-PXC_6H_{10}, 69%). 2-Methyl-1,3-butadiene is consumed mainly by the following reactions (R1–R3).

$$CH_3\text{-}2\text{-}C_4H_5\text{-}13 + H = CH_3C^{*}CH_2 + C_2H_4 \quad (R1, 37\%)$$

$$CH_3\text{-}2\text{-}C_4H_5\text{-}13 + H = PAXCH_2\text{-}2\text{-}C_4H_5 + H_2 \quad (R2, 26\%)$$

$$CH_3\text{-}2\text{-}C_4H_5\text{-}13 + H = C_3H_6 + C_2H_3 \quad (R3, 26\%)$$

As for 1,3-pentadiene, it is produced from the decomposition of hept-5-en-1-yl radical (PX7-2C_7H_{13}, 64%) and 3-methyl-hex-5-en-1-yl radical (CH_3-4-PXC_6H_{10}, 34%), and is consumed mainly by the following three reactions (R4–R6).

$$C_5H_8\text{-}13 + H = C_2H_4 + CH_3CHCH^{*} \quad (R4, 64\%)$$

$$C_5H_8\text{-}13 + H = lC_5H_7 + H_2 \quad (R5, 25\%)$$

$$C_5H_8\text{-}13 + CH_3 = lC_5H_7 + CH_4 \quad (R6, 7\%)$$

(R5) and (R6) contribute ~17% of the lC_5H_7 formation, while the following (R7) and (R8) produce most of the remaining lC_5H_7.

$$iC_4H_5 + CH_3 = lC_5H_7 + H \quad (R7, 48\%)$$

$$lC_5H_7 + CH_3 = C_6H_{10}\text{-}13 \quad (R8, 30\%)$$

lC_5H_7 dominantly decomposes to cyclopentadiene (C_5H_6), as shown in Fig. 7d.

Another important diene is 1,3-butadiene with very high concentration (up to 8.2E-3, shown in Fig. 7f), which is slightly higher than that in cyclohexane pyrolysis [9]. In cyclohexane pyrolysis, it is dominantly produced from hex-5-en-1-yl radical (PXC_6H_{11}). However, in MCH pyrolysis, 1,3-butadiene is formed by various channels such as hex-5-en-1-yl radical (42%), hept-1-en-6-yl radical (SXC_7H_{13}, 15%), 2-methyl-hex-5-en-1-yl radical (PXCH_2-5-1C_6H_{11}, 10%) and hept-5-en-1-yl radical (PX7-2C_7H_{13}, 6%). Most of 1,3-butadiene (56%) converts to ethylene and C_2H_3 by reacting with H atom, while about 39% of it is consumed by radical attack to form nC_4H_5 and iC_4H_5, which are the dominant precursors for vinylacetylene (C_4H_4), as shown in Fig. 7g.

Figure 7h–l display the mole fraction profiles of the C3 intermediates detected in this work. Propene is mainly produced from the decomposition of hept-1-en-6-yl radical (SXC_7H_{13}, 30%), the H-loss of iC_3H_7 (24%) generated from the 2-methyl-hex-5-en-1-yl radical (PXCH_2-5-1C_6H_{11}), and the association of C_2H_3 with CH_3 (14%). It is mainly consumed by reacting with H atom to produce C_2H_4 and CH_3 (72%). Allyl radical (aC_3H_5) is mainly formed from the following three reactions (R9–R11).

$$PXC_5H_9 = C_2H_4 + aC_3H_5 \quad (R9, 28\%)$$

$$C_6H_{10}\text{-}15 = aC_3H_5 + aC_3H_5 \quad (R10, 19\%)$$

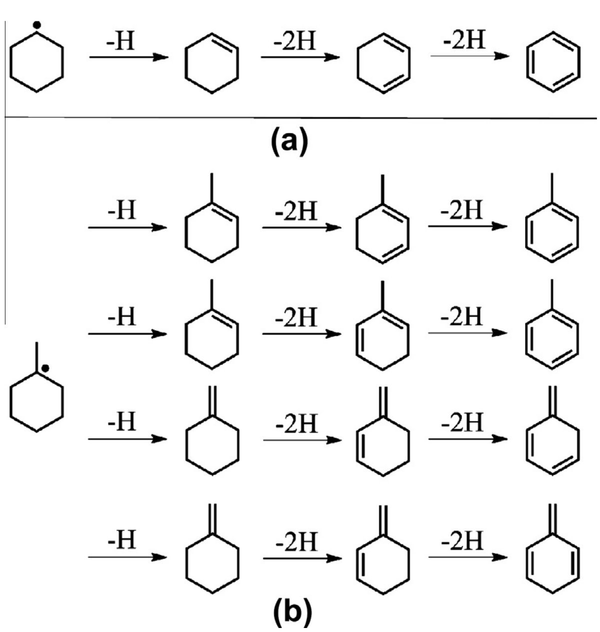
$C_7H_{14}\text{-}1 = pC_4H_9 + aC_3H_5 \qquad (R11, 15\%)$

The maximum concentration of aC_3H_5 is much lower than that in cyclohexane pyrolysis [9], where it is dominantly formed from the dissociation of 1-hexene, the initial isomerization product of cyclohexane pyrolysis. In MCH pyrolysis, its initial isomerization product such as 1-heptene (C_7H_{14}-1) also dissociates into aC_3H_5 (R11). However, the contribution of (R11) to aC_3H_5 formation is much less than that of 1-hexene decomposition since 1-heptene is not the dominant initial product of MCH pyrolysis [62]. This is a possible explanation for the lower aC_3H_5 concentration in MCH pyrolysis.

In a word, the ring opening of the cyclohexyl and cyclic C_7H_{13} radicals produces alkenyl C_6H_{11} and C_7H_{13} radicals, which are precursors of dienes (e.g. 1,3-butadiene, 1,3-pentadiene, 2-methyl-1,3-butadiene and 1,3-hexadiene), alkenes (e.g. propene and ethylene), and some radicals (such as C_4H_7, C_3H_7 and C_2H_5). Further reactions of them constitute the various intermediates detected in this work. Compared with cyclohexane pyrolysis, large concentrations of C5 (1,3-pentadiene and 2-methyl-1,3-butadiene) and C6 (1,3-hexadiene) dienes were detected. On the other hand, the maximum concentration of allyl radical is lower than that in cyclohexane pyrolysis. These differences indicate different pyrolysis chemistries between cyclohexane and MCH. In general, the present model predicts satisfactorily the pyrolysis of MCH, as shown in Figs. 3 and 7. Similarly, the prediction of these C0–C6 species by Orme et al. model [26] at 760 Torr is also satisfactory.

5.1.2. H- and/or CH_3-loss of the cyclohexyl and cyclic C_7H_{13} radicals

On the other hand, a small amount of cyclohexyl and cyclic C_7H_{13} radicals undergo stepwise dehydrogenation and/or dealkylation to produce cyclic compounds. This process is shown in Scheme 1 taken cyclohexyl and 1-methyl-cyclohexyl radicals as an example. In the experimental measurements, various cyclic intermediates were detected and quantified, including cyclohexene (cC_6H_{10}), 1,3-cyclohexadiene (cC_6H_8-13), benzene (A1), C_7H_{12}, C_7H_{10}, *o*-isotoluene and *p*-isotoluene (cC_7H_8), and toluene ($A1CH_3$). The adiabatic ionization energies (IEs) of the most probable isomers of these cyclic C_7H_{12}, C_7H_{10} and C_7H_8 species are shown in supplementary, which are taken from NIST [76] and our theoretical calculation at CBS-QB3. The measured photoionization efficiency (PIE) spectra of *m/z* 78, 92, 94 and 96 are shown in supplementary, where the structures of the most probable isomers are shown. For *m/z* 92 in Fig. S9b, the onset at 8.84 eV is in accord with the IE of toluene. There exists another onset at approximately 7.96 eV. The calculated IE of *o*-isotoluene is 7.81 eV, which is slightly lower than the NIST value (7.9 eV). The calculated IE of *p*-isotoluene is 8.11 eV. These two species are the most probable isomers in MCH pyrolysis although the measured onset is not so consistent with the calculated IEs. In our previous work on the pyrolysis of cyclohexane [9], fulvene, benzene, 1,3-cyclohexadiene and toluene were also detected. The absence of cyclic C_7H_{12} and C_7H_{10} in cyclohexane pyrolysis indicates that these intermediates come directly from the decomposition of MCH, since the detected C1–C6 intermediates in the two fuels pyrolysis are quite similar.



Scheme 1. Stepwise dehydrogenation of cyclohexyl and 1-methyl-cyclohexyl radicals. The H-loss processes include both direct H-elimination and H-abstraction via radicals attack.

The measured and simulated mole fraction profiles of these species are shown in Fig. 8. For C_7H_{10} and C_7H_{12}, their measured mole fractions include all the probable isomers shown in Figs. S9c and S9d since they were not separable experimentally. Similarly, the simulated mole fractions of C_7H_{12} and C_7H_{10} also include all the probable isomers. Our simulation shows that the major constitution of C_7H_{12} is 4-methyl-cyclohexene while the contribution of 1,3-heptadiene and 5-methyl-1,3-hexadiene is insignificant under the three pressures. For C_7H_{10}, the contribution of 1,3,5-heptatriene increases with pressure, which constitutes about 70% of the total mole fraction at 760 Torr. Reaction flux analysis shows that 1,3,5-heptatriene mainly comes from the reaction of 1,3-pentadiene with vinyl radical. For cC_7H_8, its two probable isomers are included in the experimental measurement and simulation. The model predicts satisfactorily the mole fractions of these intermediates except for that of cC_7H_8. According to the reaction flux analysis, cC_7H_8 is formed from the stepwise dehydrogenation of

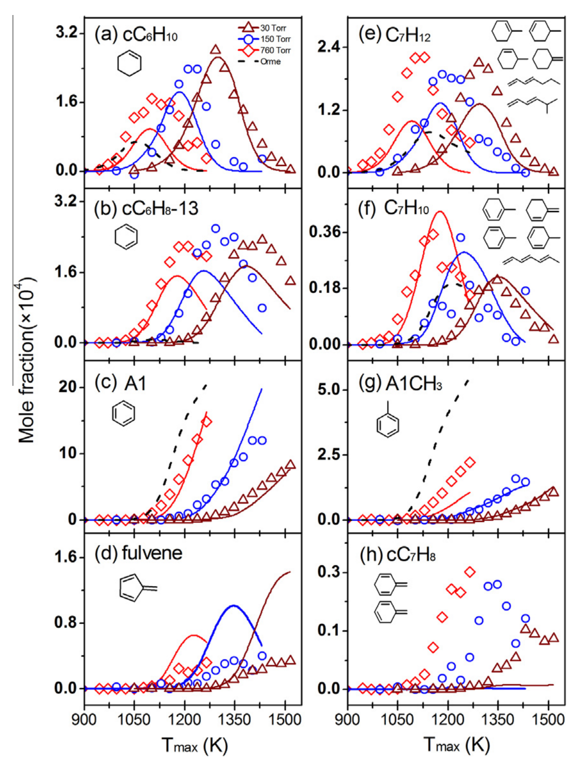


Fig. 8. Measured (symbols) and simulated (lines) mole fraction profiles of C6-C7 products in MCH pyrolysis at 30, 150 and 760 Torr. Solid lines are the simulated results with the present model, while dash lines are the simulated results by Orme et al. [26] model at 760 Torr.

1-methyl-cyclohexyl ($CH_3TXcC_6H_{10}$) and $PXCH_2cC_6H_{11}$ radicals, and mainly undergoes H-loss to form benzyl radical ($A1CH_2$). The discrepancy between experiment and simulation indicates the corresponding mechanism needs further development.

Benzene and toluene are finally formed via the stepwise dehydrogenation and/or dealkylation of the cyclohexyl and cyclic C_7H_{13} radicals. It is noticeable that the maximum mole fractions of benzene are much higher than those of toluene in the three pressure experiments. Similarly, the maximum mole fractions of 1,3-cyclohexadiene are also much higher than those of the cyclic C_7H_{10}. The reaction pathways for benzene and toluene formation from cyclohexyl and cyclic C_7H_{13} radicals are shown in Fig. 9. It can be seen that the stepwise dehydrogenation of 1-methyl-cyclohexyl radical ($CH_3TXcC_6H_{10}$) and 2-methyl-cyclohexyl radical ($CH_3S2XcC_6H_{10}$) through 1-methyl-cyclohexene (CH_3-1-cC_6H_9), 1-methyl-1,3-cyclohexadiene (CH_3-1-cC_6H_7-13) and 2-methyl-1,3-cyclohexadiene (CH_3-2-cC_6H_7-13) forms toluene. On the other hand, the stepwise dehydrogenation and/or dealkylation of 2-methyl-cyclohexyl ($CH_3S2XcC_6H_{10}$), 3-methyl-cyclohexyl ($CH_3S3XcC_6H_{10}$) and 4-methyl-cyclohexyl ($CH_3S4XcC_6H_{10}$) radicals produce benzene through cyclohexene (cC_6H_{10}), 3-methyl-cyclohexene (CH_3-3-cC_6H_9), 4-methyl-cyclohexene (CH_3-4-cC_6H_9), 5-methyl-1,3-cyclohexadiene (CH_3-5-cC_6H_7-13) and 1,3-cyclohexadiene (cC_6H_8-13). Moreover, the stepwise dehydrogenation of cyclohexyl radical (cC_6H_{11}) produced from the CH_3-loss of MCH is another source of benzene. Thus, much more carbon flux flows into C6-ring products than the C7-ring products. This explains the higher mole fraction of 1,3-cyclohexadiene and benzene than cyclic C_7H_{10} and toluene.

Furthermore, the reaction flux analysis of toluene and benzene at 30 and 760 Torr was performed when 80% of MCH is consumed, as shown in Fig. 10a and b. Toluene is dominantly produced from the five cyclic C_7H_9 radicals, which mainly come from the stepwise dehydrogenation of 1-methyl-cyclohexyl radical ($CH_3TXcC_6H_{10}$) and 2-methyl-cyclohexyl radical ($CH_3S2XcC_6H_{10}$). Among them, the 3-methyl-1,3-cyclohexadienyl (CH_3-3-$SAXcC_6H_6$) radical has the major contribution. Besides, the association of benzyl radical with H atom also has some contribution for toluene formation. These observations are different to that in cyclohexane pyrolysis [9], where toluene ($A1CH_3$) is mainly produced from the self-combination of propargyl radical (C_3H_3), through the following three reactions via phenyl (A1-) and benzyl radical ($A1CH_2$).

$$C_3H_3 + C_3H_3 = A1- + H \quad (R12)$$

$$A1CH_2 + H = A1- + CH_3 \quad (R13)$$

$$A1CH_2 + H = A1CH_3 \quad (R14)$$

For the simulation results of C_7H_{12} by Orme et al. model [26], all the probable isomers shown in Fig. S9d are included. These four cyclic C_7H_{12} compounds are also produced from the H-loss of the cyclic C_7H_{13} radicals. But, there are no further reactions of these cyclic C_7H_{12} compounds in Orme et al. model [26]. The simulated mole fraction of C_7H_{12} is close to that in our model, but it might

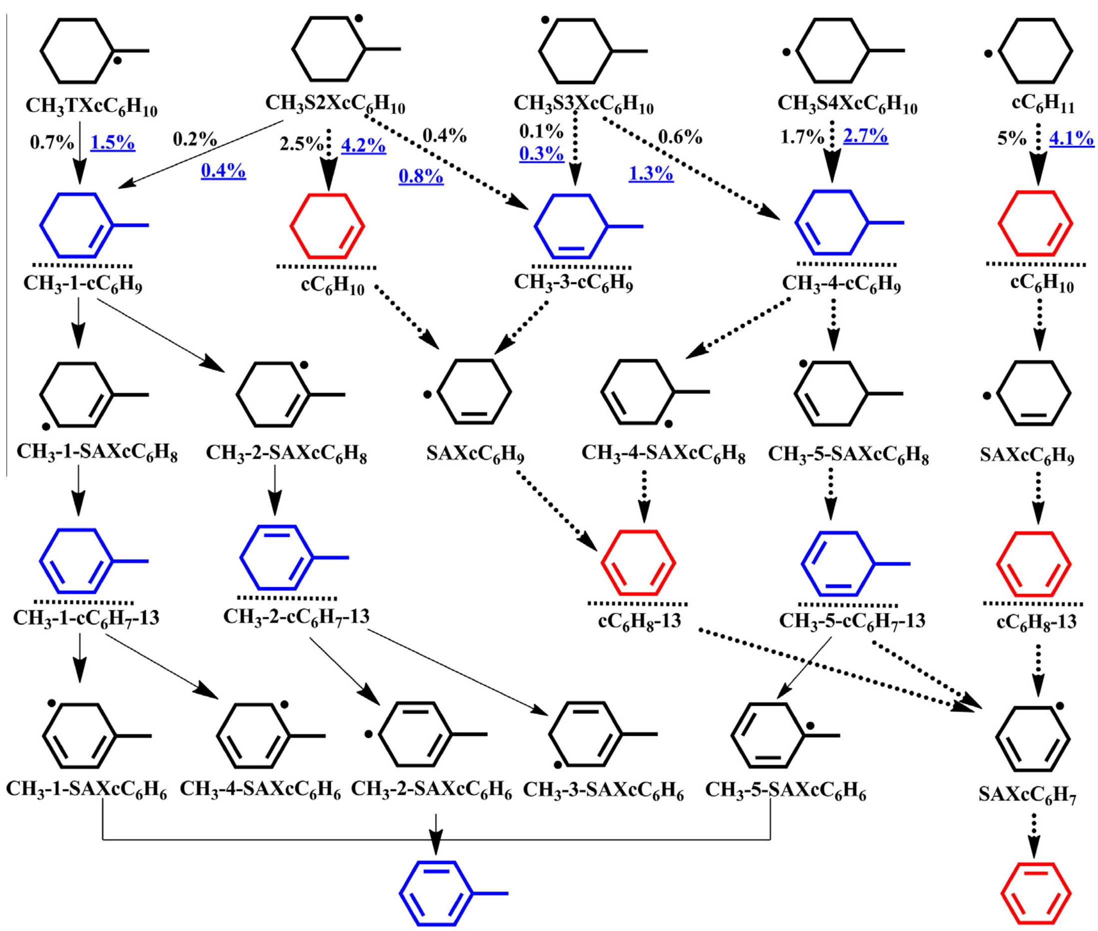


Fig. 9. Diagram of the stepwise dehydrogenation and/or dealkylation of cyclohexyl and cyclic C_7H_{13} radicals in MCH pyrolysis at 30 and 760 Torr. The conversion of MCH is approximately 80%. Only important pathways are shown. The numbers indicate the carbon flux of the pathway divided by total carbon flux of MCH consumption. Black for 30 Torr while blue and underlines for 760 Torr. Species labeled with color and underlines were detected in the experiment. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

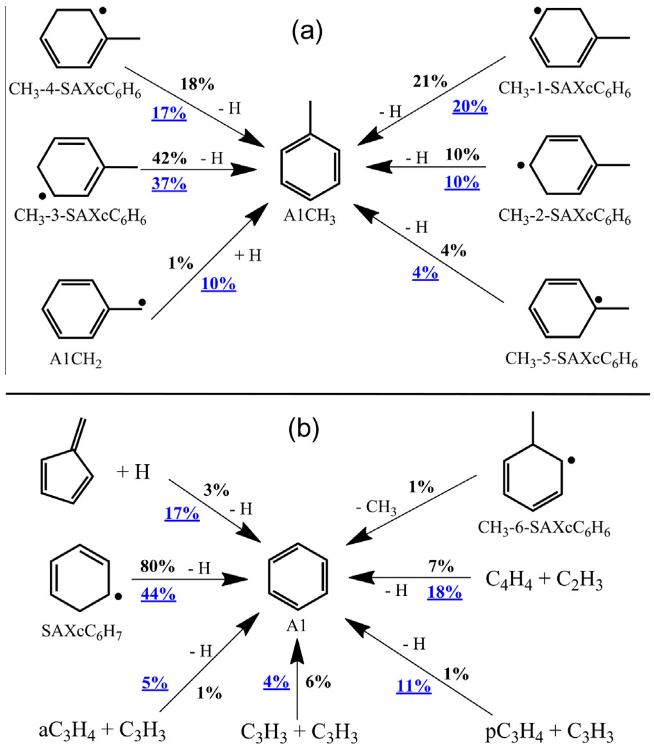


Fig. 10. Reaction flux analysis of toluene and benzene formation in MCH pyrolysis at 30 Torr (in black) and 760 Torr (in blue and underlines). The conversion of MCH at both pressures is approximately 80%. The numbers indicate the contribution percentages. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

be much lower if the further reactions of these species are included. For the simulation result of C_7H_{10}, only 1,3,5-heptatriene is included since no cyclic C_7H_{10} can be produced. Thus toluene is not originated from the cyclic C_7H_{13} radicals in Orme et al. model [26]. Instead, the reaction of C_3H_3 with 1,3-butadiene dominates toluene formation (R15). As shown in Fig. 8g, the mole fraction of toluene is over-predicted when we used Orme et al. model to simulate the present MCH pyrolysis experiment.

$$C_3H_3 + C_4H_6 = A1CH_3 + H \qquad (R15, 95\%)$$

Multiple formation channels of benzene are observed in MCH pyrolysis (shown in Fig. 10b), which is similar to that in cyclohexane pyrolysis [9]. In 30 Torr pyrolysis, the H-loss of the cyclohexadienyl ($SAXcC_6H_7$) radical contributes the most, which is produced from the stepwise dehydrogenation of the cyclohexyl radical and the decomposition of 2-methyl-cyclohexyl ($CH_3S2XcC_6H_{10}$), 3-methyl-cyclohexyl ($CH_3S3XcC_6H_{10}$) and 4-methyl-cyclohexyl (CH_3-$S4XcC_6H_{10}$) radicals, as shown in Fig. 9. Around 4% of benzene is produced from the CH_3-loss of the 6-methyl-1,3-cyclohexadienyl radical (CH_3-6-$SAXcC_6H_6$) and the H-assisted isomerization of fulvene which is mainly produced by the following two reactions.

$$C_3H_3 + aC_3H_5 = \text{fulvene} + 2H \qquad (R16, 78\%)$$

$$C_3H_3 + C_3H_3 = \text{fulvene} + H \qquad (R17, 17\%)$$

The reactions of small resonantly stabilized radicals, such as $C_4H_4 + C_2H_3$, $C_3H_3 + C_3H_3$, $C_3H_3 + aC_3H_4$, and $C_3H_3 + pC_3H_4$ contribute about 15% of benzene formation. Comparing the results at 30 and 760 Torr in Fig. 10b, the contribution from H-loss of the cyclohexadienyl radical decreases with increasing pressure, while that from the H-assisted isomerization of fulvene and the reactions of small resonantly stabilized radicals increases with pressure. Under 760 Torr pyrolysis, the H-assisted isomerization of fulvene and the reactions of small resonantly stabilized radicals produce much more benzene, which is probable another reason for the much higher mole fraction of benzene than toluene.

The simulated mole fraction of benzene (A1) by Orme et al. model [26] agrees with the experimental result while that of 1,3-cyclohexadiene is significantly underpredicted. Reaction flux analysis shows that benzene is mainly produced by cyclohexadienyl radical which comes from the combination of nC_4H_5 with C_2H_2. Another important channel is the reaction of 1,3-butadiene with vinyl radical.

$$C_2H_3 + C_4H_6 = A1 + H_2 + H \qquad (R18)$$

Thus, the benzene formation pathways in Orme et al. model [26] are also different to those in our current model.

5.2. Premixed MCH flames

The temperature profile and the measured and simulated mole fraction profiles of major species of the studied flame are displayed in Fig. 11. For the simulation of the premixed flame, the measured temperature profile was used as input values. The calculation used mixture-average transport including Soret diffusion. The converged solution gradient and curvature is 0.19 and 0.39, respectively. The mole fraction of Ar decreases from the burner surface due to the mole expansion effect, then becomes almost constant after the reaction zone. MCH is consumed at approximately 8.0 mm while the residual oxygen is consumed by the combustion intermediates after the complete consumption of the fuel. The concentrations of incomplete oxidation products such as CO and H_2 are very high at the postflame zone because of the rich flame condition. In general, the simulated results match the mole expansion effect, the consumption of MCH and O_2, and the formation of H_2, H_2O, CO and CO_2, providing a preliminary validation of the kinetic model under oxidization circumstance.

The C1–C7 intermediates with the measured and simulated mole fraction profiles are shown in Fig. 12. It should be noted that there exists large discrepancy between experiment and simulation close to the burner surface, which is mainly caused by the perturbation of the sampling nozzle [77–79]. Thus the measured data at distances shorter than 2 mm from the burner surface might not represent the true mole fractions of the supposedly measured species. The reaction flux analysis of MCH is shown in supplementary. According to the reaction flux analysis, the initial decomposition and isomerization of MCH are not important as those in 30 Torr pyrolysis. Instead, MCH is dominantly consumed by H-abstraction via radical attack forming the cyclic C_7H_{13} radicals, among which

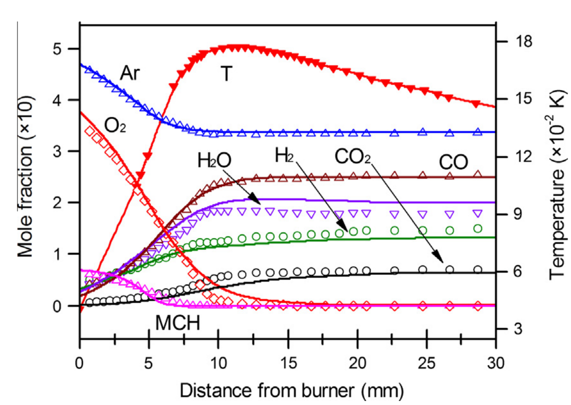


Fig. 11. Measured (open symbols) and simulated (solid lines) mole fraction profiles of major species in the premixed flame with equivalence ratio of 1.75 with the measured temperature profile (solid symbol and line).

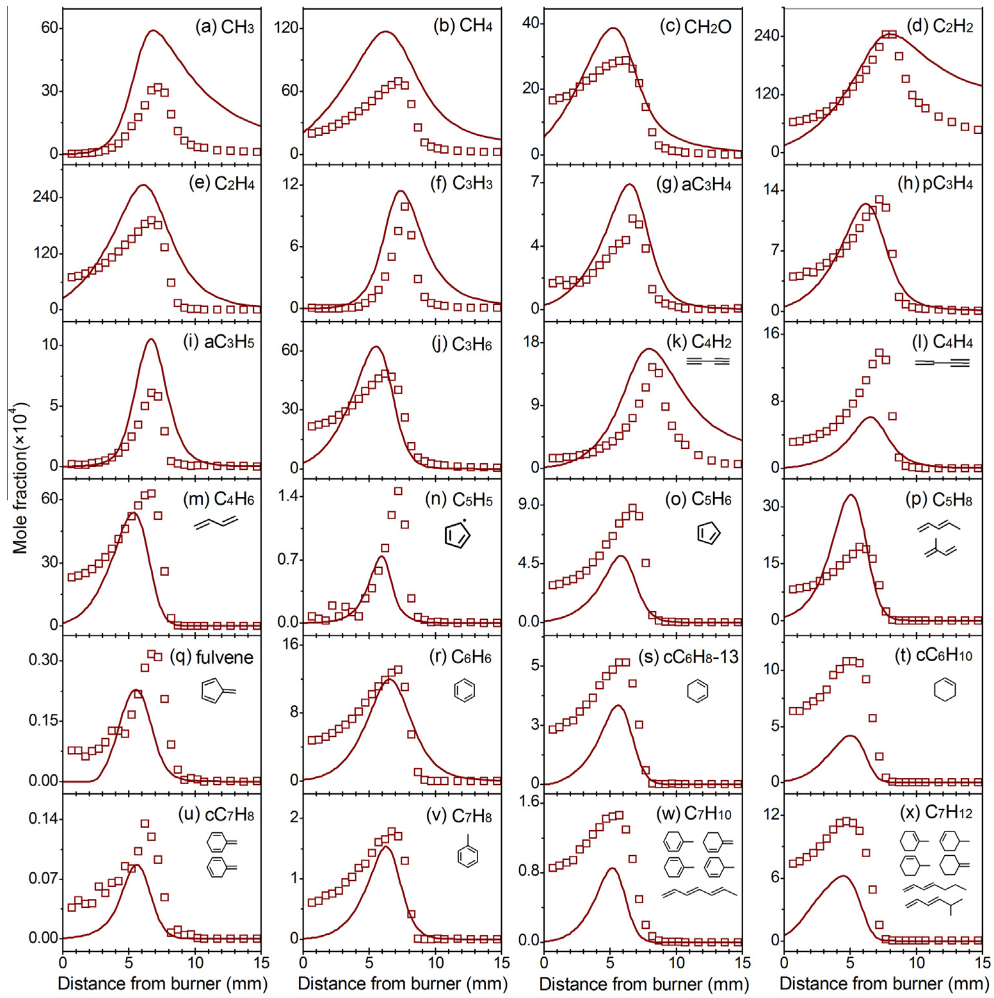


Fig. 12. Measured (open symbols) and simulated (solid lines) mole fraction profiles of the C1–C7 intermediates in the premixed flame with equivalence ratio of 1.75.

the channels yielding 3-methyl-cyclohexyl ($CH_3S3XcC_6H_{10}$) and 2-methyl-cyclohexyl ($CH_3S2XcC_6H_{10}$) radicals are more favored. The cyclic C_7H_{13} radicals are dominantly isomerized to alkenyl C_7H_{13} radicals via the ring-opening reactions, as described in the MCH pyrolysis. The alkenyl C_7H_{13} radicals are then decomposed into various intermediates. The reaction flux analysis of cyclohexyl and cyclic C_7H_{13} radicals is shown in supplementary. The following discussion will focus on some key intermediates for illustration.

Figure 12m shows that the relative concentration of 1,3-butadiene is very high in the studied flame, in agreement with the previous studies [20,80]. Reaction flux analysis reveals that most of the 1,3-butadiene is produced from the alkenyl C_7H_{13} radicals such as 2-methyl-hex-5-en-1-yl radical ($PXCH_2$-5-$1C_6H_{11}$, 18%), hept-1-en-6-yl radical (SXC_7H_{13}, 21%), hept-5-en-1-yl radical (PX7-$2C_7H_{13}$, 11%) and hept-6-en-1-yl radical (PXC_7H_{13}, 8%). The contribution of hex-5-en-1-yl radical (PXC_6H_{11}, 15%) is not as important as that in MCH pyrolysis since the decomposition of MCH via CH_3-loss is unimportant in the flame. Figure 12p shows the mole fraction of C_5H_8, including both 2-methyl-1,3-butadiene and 1,3-pentadiene. Similar to the MCH pyrolysis, these two species are formed directly from the alkenyl C_7H_{13} radicals. These two species were also detected and quantified in the study by Skeen et al. [8], according to which the concentration of 2-methyl-1,3-butadiene is slightly higher than 1,3-pentadiene.

Similar to the MCH pyrolysis, the cyclic C6–C7 species were also detected in the flame, as shown in Fig. 12q–x with the measured and simulated mole fraction profiles. The measured PIE curves of these species are similar to those in MCH pyrolysis, and are also similar to the study by Skeen et al. [8]. The simulated mole fractions of C_7H_{10} and C_7H_{12} include all the probable isomers shown in Figs. S9c and S9d. Besides toluene, cC_7H_8 (*o*-isotoluene and *p*-isotoluene) was also detected. The measured IE onset is about 7.91 eV, which is close to that of *o*-isotoluene. Different from the prediction in MCH pyrolysis, the cC_7H_8 formed from the stepwise dehydrogenation of 1-methyl-cyclohexyl ($CH_3TXcC_6H_{10}$) and $PXCH_2cC_6H_{11}$ (⌬) radicals is predicted satisfactorily. For fulvene, the prediction agrees with the experimental result, where it is mainly produced from (R16) and (R17). Under flame condition, the measured mole fractions of 1,3-cyclohexadiene and benzene are also higher than those of cyclic C_7H_{10} and toluene. The possible explanation is similar to that in MCH pyrolysis, as shown in Fig. S12. The formations of toluene and benzene will be discussed below.

The reaction flux analysis in Fig. S13 shows that toluene is dominantly formed by the H-loss of the five C_7H_9 radicals coming from the stepwise dehydrogenation of MCH, which is similar to MCH pyrolysis. Benzene is formed mainly through the H-loss of cyclohexadienyl radical ($SAXcC_6H_7$, 75%), the CH_3 displacement by H atom in toluene (7%), and the H-assisted isomerization of fulvene (10%) under the flame condition. Compared to the formation pathways under the pyrolytic condition, the contribution from the reactions of small resonantly stabilized radicals is much lower.

The sensitivity analysis of toluene and benzene is presented in Fig. 13, calculated at 6.3 mm from the burner surface for toluene and 6.6 mm for benzene. The formations of toluene and benzene are sensitive to reactions related to the C7- and C6-cyclic intermediates. For example, the H-loss of 1-methyl-cyclohexyl, 2-methyl-cyclohexyl and 4-methyl-cyclohexyl radicals forming methyl-cyclohexenes has positive effect on toluene formation, while their ring-opening isomerization has negative impact. Besides, the ring-opening decomposition of methyl-cyclohexenes such as 1-methyl-cyclohexene and 4-methyl-cyclohexene also has negative effect on toluene formation. According to the reaction flux analysis, toluene is dominantly consumed by H-abstraction to form benzyl radical, which further produces toluene via obtaining a H atom. This explains the large positive effect of the combination of benzyl radical with H atom on toluene formation, while the H-abstraction of toluene forming benzyl radical has large negative effect. For benzene formation, the CH_3-loss of 2-methyl-cyclohexyl radical has the largest positive impact while its ring-opening isomerization has very large negative effect. The H-loss of 4-methyl-cyclohexyl, 2-methyl-cyclohexyl and 3-methyl-cyclohexyl radicals forming methyl-cyclohexenes also has positive effect on benzene formation, while their ring-opening isomerization has negative impact. Different to the sensitivity analysis of toluene, the reactions of 1-methyl-cyclohexyl radical have little effect on benzene formation. Besides these C7-ring channels, the reactions of C6-ring intermediates such as cyclohexyl radical, cyclohexene and 1,3-cyclohexadiene also have effect on benzene formation.

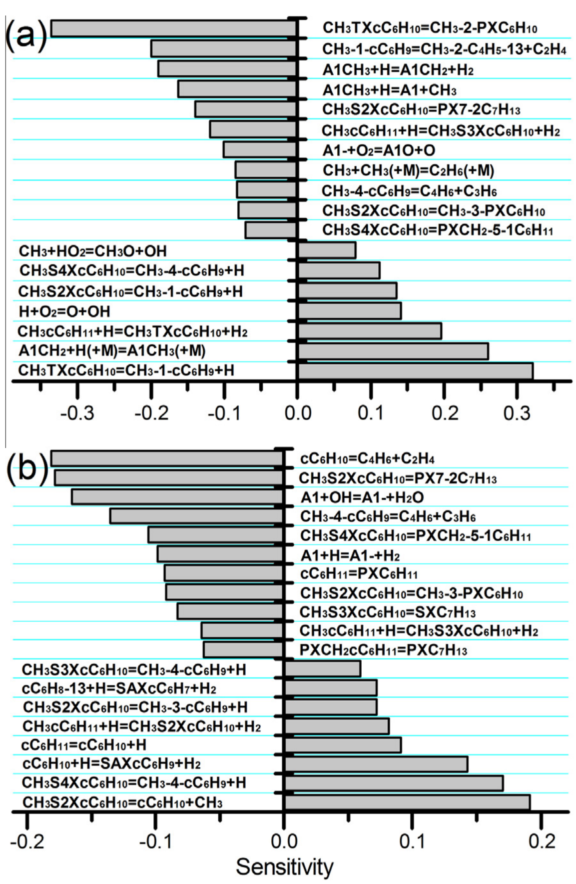


Fig. 13. Normalized sensitivity analysis of toluene (a) and benzene (b) in the premixed flame (ϕ = 1.75) calculated at 6.3 mm from burner surface for toluene and 6.6 mm for benzene. Reactions with absolute sensitivity coefficients larger than 0.07 and 0.06 are considered for toluene and benzene, respectively.

Furthermore, the present kinetic model was used to simulate the experimental data of MCH premixed flames reported by Skeen et al. [8] with equivalence ratios of 1.0, 1.75 and 1.9 measured at 10, 15 and 30 Torr, respectively. In Skeen et al. work [8], the experimental method used is similar to our work, which allows the identification of species via precise energy scan. Stoichiometric and rich flames were chosen to study the fuel consumption and aromatics formation under oxidation and pyrolytic conditions. The previous study provides valuable information on MCH consumption and aromatics formation in MCH flames.

The comparisons of major products and some intermediates are shown in supplementary. For the major products, the simulation result by our model is satisfactory. As discussed above and also in Skeen et al. work [8], MCH is consumed by three kinds of reactions: unimolecular decomposition, isomerization and H-abstraction. According to the reaction flux analysis of our simulation, about 97% of MCH is consumed via H-abstraction through H atom, OH radical and O atom attack under stoichiometric flame. At fuel rich flames with ϕ = 1.75 and 1.9, the contribution of unimolecular decomposition and isomerization increases, which is similar to that of the flame studied in this work.

For intermediates shown in Fig. S15, the simulated mole fractions of most species are also satisfactory. The biggest discrepancies are 2-methyl-1,3-butadiene and 1,3-pentadiene in the ϕ = 1.75 flame, and cyclohexene and toluene in ϕ = 1.0 and ϕ = 1.9 flames. These discrepancies indicate the current model needs further improvement. Nevertheless, the experimental uncertainty cannot be ruled out, as reported by Skeen et al. [8], the experimental uncertainty in the mole fraction profiles can be as large as a factor of 2. In Skeen et al. work [8], they concluded that the sequential H-abstraction of cyclohexyl radical is the dominant channel for benzene formation. Our simulation shows that the benzene

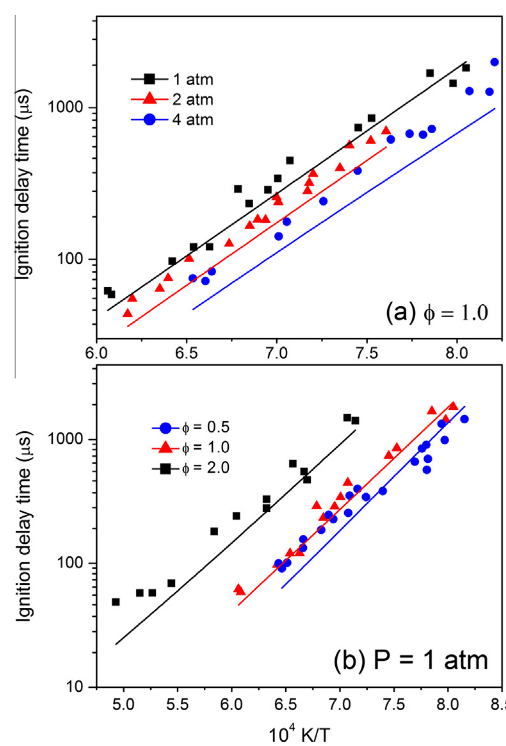


Fig. 14. (a) Ignition delay times for a 1.0% MCH in Ar mixture, ϕ = 1.0 at 1, 2 and 4 atm. (b) Ignition delay times for MCH in Ar mixture. P = 1 atm at ϕ = 0.5 (0.5% MCH, 10.5% O_2), 1.0 (1.0% MCH, 10.5% O_2) and 2.0 (1.0% MCH, 5.25% O_2). Symbols are experimental results by Orme et al. [26]; lines are modeling predictions in this work.

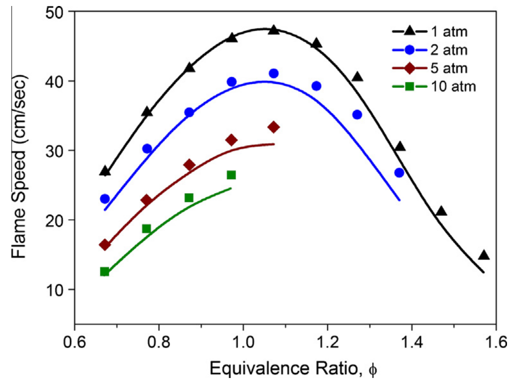


Fig. 15. Laminar flame speeds of MCH at various pressures with air as the oxidizer, unburned gas temperature of 353 K. Symbols are experimental results by Wu et al. [32]; lines are modeling predictions in this work.

formation channels are complicated, the sequential H-abstraction of cyclohexyl radical is one of the important, but not the dominant channel, especially in the ϕ = 1.0 flame.

5.3. Ignition delay times and flame speeds of MCH

The global properties of MCH combustion such as ignition delay times and laminar flame speeds were chosen to get a further validation of the present kinetic model. The ignition delay times were measured by Orme et al. [26] at various pressures and equivalence ratios, as shown in Fig. 14. The simulation was performed by Chemkin-PRO software [72] with the Closed Homogenous Batch Reactor. The present model reproduces satisfactorily the ignition delay times of MCH at various experimental conditions. Figure 15 presents the laminar flame speeds of MCH at 1, 2, 5 and 10 atm as a function of equivalence ratios measured by Wu et al. [32]. They declared the uncertainty of the measured flame speeds is ±2 cm/s. The simulation was performed with the PREMIX code in Chemkin-PRO software. All calculations used mixture-average transport including Soret diffusion. Considering the experimental uncertainty, the simulated flame speeds are in agreement with the measurement.

6. Conclusions

The pyrolysis of MCH from low to atmospheric pressures, the low pressure premixed MCH flame with equivalence ratio of 1.75 were studied using the SVUV-PIMS technique. A kinetic model with 249 species and 1570 reactions was built based on the newly developed sub-mechanism of MCH, and validated by the pyrolysis and flame data. The model reproduces the mole fraction profiles of most pyrolysis and flame species. A large amount of 1,3-butadiene is formed under both pyrolytic and flame conditions. High concentrations of benzene and toluene are detected, which mainly come from the decomposition of cyclohexyl and cyclic C_7H_{13} radicals. The propensity of benzene formation is larger than toluene. The H-assisted isomerization of fulvene and the reactions of small resonantly stabilized radicals such as through the $C_4 + C_2$, $C_3 + C_3$ channels also have contribution to benzene formation. Special attention should be paid to MCH combustion considering the large formation of 1,3-butadiene, benzene and toluene. Furthermore, the present kinetic model was validated by speciation in premixed flames, the ignition delay times and flame speeds of MCH reported in the literature, and satisfactory agreement is observed.

Acknowledgments

Authors are grateful for the reviewers' insightful suggestions and funding supports from National Basic Research Program of China (973 Program) (2013CB834602), Natural Science Foundation of China (50925623 and U1232127), and Chinese Academy of Sciences. We appreciate Dr. Jiuzhong Yang and Mr. Hanfeng Jin in flame experiment measurement, and Professor Katharina Kohse-Höinghaus and Dr. Yuyang Li for many helpful comments to improve this manuscript. The Super Computing Center of USTC is appreciated for Gaussian calculation.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.combustflame.2013.08.011.

References

[1] J. Guthrie, P. Fowler, R. Sabourin, Gasoline and Diesel, Fuel Survey, 2003.
[2] Northrop Grumman, Diesel Fuel Oils, 2003, Report NGMS-232 PPS, January 2004.
[3] P.J. Bennett, D. Gregory, R.A. Jackson, Combust. Sci. Technol. 115 (1996) 83–103.
[4] D. Voisin, A. Marchal, M. Reuillon, J.C. Boettner, M. Cathonnet, Combust. Sci. Technol. 138 (1998) 137–158.
[5] O. Lemaire, M. Ribaucour, M. Carlier, R. Minetti, Combust. Flame 127 (2001) 1971–1980.
[6] M.E. Law, P.R. Westmoreland, T.A. Cool, J. Wang, N. Hansen, C.A. Taatjes, T. Kasper, Proc. Combust. Inst. 31 (2007) 565–573.
[7] E.J. Silke, W.J. Pitz, C.K. Westbrook, M. Ribaucour, J. Phys. Chem. A 111 (2007) 3761–3775.
[8] S.A. Skeen, B. Yang, A.W. Jasper, W.J. Pitz, N. Hansen, Energy Fuels 25 (2011) 5611–5625.
[9] Z.D. Wang, Z.J. Cheng, W.H. Yuan, J.H. Cai, L.D. Zhang, F. Zhang, F. Qi, J. Wang, Combust. Flame 159 (2012) 2243–2253.
[10] B. Husson, O. Herbinet, P.A. Glaude, S.S. Ahmed, F. Battin-Leclerc, J. Phys. Chem. A 116 (2012) 5100–5111.
[11] A. Violi, S. Yan, E.G. Eddings, A.F. Sarofim, S. Granata, T. Faravelli, E. Ranzi, Combust. Sci. Technol. 174 (2002) 399–417.
[12] A. Agosta, N.P. Cernansky, D.L. Miller, T. Faravelli, E. Ranzi, Exp. Therm. Fluid Sci. 28 (2004) 701–708.
[13] J.A. Cooke, M. Bellucci, M.D. Smooke, A. Gomez, A. Violi, T. Faravelli, E. Ranzi, Proc. Combust. Inst. 30 (2005) 439–446.
[14] S. Humer, A. Frassoldati, S. Granata, T. Faravelli, E. Ranzi, R. Seiser, K. Seshadri, Proc. Combust. Inst. 31 (2007) 393–400.
[15] H. Bufferand, L. Tosatto, B. La Mantia, M.D. Smooke, A. Gomez, Combust. Flame 156 (2009) 1594–1603.
[16] S. Humer, R. Seiser, K. Seshadri, J. Propul. Power 27 (2011) 847–855.
[17] C. Ji, F.N. Egolfopoulos, Proc. Combust. Inst. 33 (2011) 955–961.
[18] P.H. Taylor, W.A. Rubey, Energy Fuels 2 (1988) 723–728.
[19] T.C. Brown, K.D. King, Int. J. Chem. Kinet. 21 (1989) 251–266.
[20] S. Zeppieri, K. Brezinsky, I. Glassman, Combust. Flame 108 (1997) 266–286.
[21] C.S. McEnally, L.D. Pfefferle, Proc. Combust. Inst. 30 (2005) 1425–1432.
[22] Y. Yang, A.L. Boehman, Proc. Combust. Inst. 32 (2009) 419–426.
[23] W.J. Pitz, C.V. Naik, T.N. Mhaoldúin, C.K. Westbrook, H.J. Curran, J.P. Orme, J.M. Simmie, Proc. Combust. Inst. 31 (2007) 267–275.
[24] G. Mittal, C.J. Sung, Combust. Flame 156 (2009) 1852–1855.
[25] R.D. Hawthorn, A.C. Nixon, AIAA J. 4 (1966) 513–520.
[26] J.P. Orme, H.J. Curran, J.M. Simmie, J. Phys. Chem. A 110 (2006) 114–131.
[27] S.S. Vasu, D.F. Davidson, Z. Hong, R.K. Hanson, Energy Fuels 23 (2009) 175–185.
[28] J. Vanderover, M.A. Oehlschlaeger, Int. J. Chem. Kinet. 41 (2009) 82–91.
[29] Z. Hong, K.-Y. Lam, D.F. Davidson, R.K. Hanson, Combust. Flame 158 (2011) 1456–1468.
[30] S.S. Vasu, D.F. Davidson, R.K. Hanson, Combust. Flame 156 (2009) 736–749.
[31] C.S. Ji, E. Dames, B. Sirjean, H. Wang, F.N. Egolfopoulos, Proc. Combust. Inst. 33 (2011) 971–978.
[32] F. Wu, A.P. Kelley, C.K. Law, Combust. Flame 159 (2012) 1417–1425.
[33] R. Sivaramakrishnan, J.V. Michael, Combust. Flame 156 (2009) 1126–1134.
[34] R. Bounaceur, V. Burklé-Vitzthum, P.-M. Marquaire, L. Fusetti, J. Anal. Appl. Pyrolysis 103 (2013) 240–254.
[35] F. Qi, R. Yang, B. Yang, C.Q. Huang, L.X. Wei, J. Wang, L.S. Sheng, Y.W. Zhang, Rev. Sci. Instrum. 77 (2006) 084101.
[36] J.H. Cai, L.D. Zhang, F. Zhang, Z.D. Wang, Z.J. Cheng, W.H. Yuan, F. Qi, Energy Fuels 26 (2012) 5550–5568.
[37] Y.Y. Li, F. Qi, Acc. Chem. Res. 43 (2010) 68–78.
[38] F. Qi, Proc. Combust. Inst. 34 (2013) 33–63.

[39] Z.D. Wang, A. Lucassen, L.D. Zhang, J.Z. Yang, K. Kohse-Höinghaus, F. Qi, Proc. Combust. Inst. 33 (2011) 415–423.
[40] Y.Y. Li, L.D. Zhang, Z.Y. Tian, T. Yuan, J. Wang, B. Yang, F. Qi, Energy Fuels 23 (2009) 1473–1485.
[41] J.H. Kint, Combust. Flame 14 (1970) 279–281.
[42] R.A. Shandross, J.P. Longwell, J.B. Howard, Combust. Flame 85 (1991) 282–284.
[43] R.M. Fristrom, Flame Structure and Processes, Oxford University Press, New York, 1995.
[44] A.T. Hartlieb, B. Atakan, K. Kohse-Höinghaus, Combust. Flame 121 (2000) 610–624.
[45] T.A. Cool, K. Nakajima, T.A. Mostefaoui, F. Qi, A. McIlroy, P.R. Westmoreland, M.E. Law, L. Poisson, D.S. Peterka, M. Ahmed, J. Chem. Phys. 119 (2003) 8356–8365.
[46] T.A. Cool, J. Wang, K. Nakajima, C.A. Taatjes, A. McIlroy, Int. J. Mass Spectrom. 247 (2005) 18–27.
[47] J. Wang, B. Yang, T.A. Cool, N. Hansen, T. Kasper, Int. J. Mass Spectrom. 269 (2008) 210–220.
[48] Z.Y. Zhou, M.F. Xie, Z.D. Wang, F. Qi, Rapid Commun. Mass Spectrom. 23 (2009) 3994–4002.
[49] Z.Y. Zhou, L.D. Zhang, M.F. Xie, Z.D. Wang, D.N. Chen, F. Qi, Rapid Commun. Mass Spectrom. 24 (2010) 1335–1342.
[50] M.F. Xie, Z.Y. Zhou, Z.D. Wang, D.N. Chen, F. Qi, Int. J. Mass Spectrom. 293 (2010) 28–33.
[51] M.F. Xie, Z.Y. Zhou, Z.D. Wang, D.N. Chen, F. Qi, Int. J. Mass Spectrom. 303 (2011) 137–146.
[52] B. Yang, J. Wang, T.A. Cool, N. Hansen, S. Skeen, D.L. Osborn, Int. J. Mass Spectrom. 309 (2012) 118–128.
[53] Photonionization Cross Section Database (Version 1.0), National Synchrotron Radiation Laboratory, Hefei, China, 2011, <http://flame.nsrl.ustc.edu.cn/en/database.htm>.
[54] N. Hansen, S.J. Klippenstein, J.A. Miller, J. Wang, T.A. Cool, M.E. Law, P.R. Westmoreland, T. Kasper, K. Kohse-Höinghaus, J. Phys. Chem. A 110 (2006) 4376–4388.
[55] S. Soorkia, A.J. Trevitt, T.M. Selby, D.L. Osborn, C.A. Taatjes, K.R. Wilson, S.R. Leone, J. Phys. Chem. A 114 (2010) 3340–3354.
[56] K. Raghavachari, G.W. Trucks, J.A. Pople, M. Head-Gordon, Chem. Phys. Lett. 157 (1989) 479–483.
[57] J.A. Montgomery, J.W. Ochterski, G.A. Petersson, J. Chem. Phys. 101 (1994) 5900–5909.
[58] M.J. Frisch et al., Gaussian 09, Revision B.01, Gaussian, Inc., Wallingford, CT, 2010.
[59] V. Mokrushin, V. Bedanov, W. Tsang, M. Zachariah, V. Knyazev, ChemRate, Version 1.5.8, National Institute of Standard and Technology, Gaithersburg, MD, 2009.
[60] H. Wang, E. Dames, B. Sirjean, D.A. Sheen, R. Tangko, A. Violi, J.Y.W. Lai, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, R.P. Lindstedt, A High-temperature Chemical Kinetic Model of n-alkane (up to n-dodecane), Cyclohexane, and Methyl-, Ethyl-, n-Propyl and n-Butyl-Cyclohexane Oxidation at High Temperatures, JetSurF version 2.0, September 19, 2010, <http://melchior.usc.edu/JetSurF/JetSurF2.0>.
[61] G. da Silva, J.W. Bozzelli, J. Phys. Chem. A 112 (2008) 3566–3575.
[62] F. Zhang, Z.D. Wang, Z.H. Wang, L.D. Zhang, Y.Y. Li, F. Qi, Energy Fuels 27 (2013) 1679–1687.
[63] B. Sirjean, P.A. Glaude, M.F. Ruiz-Lopèz, R. Fournet, J. Phys. Chem. A 112 (2008) 11598–11610.
[64] A.M. Knepp, G. Meloni, L.E. Jusinski, C.A. Taatjes, C. Cavallotti, S.J. Klippenstein, Phys. Chem. Chem. Phys. 9 (2007) 4315–4331.
[65] I. Iwan, W.S. McGivern, J.A. Manion, W. Tsang, The decomposition and isomerization of cyclohexyl and 1-hexenyl radicals, in: Proceedings of 5th US Combustion Meeting, San Diego, CA 2007, CO2.
[66] J.H. Kiefer, K.S. Gupte, L.B. Harding, S.J. Klippenstein, J. Phys. Chem. A 113 (2009) 13570–13583.
[67] S.G. Davis, C.K. Law, H. Wang, J. Phys. Chem. A 103 (1999) 5889–5899.
[68] H. Wang, M. Frenklach, Combust. Flame 110 (1997) 173–221.
[69] L.V. Moskaleva, M.C. Lin, J. Comput. Chem. 21 (2000) 415–425.
[70] S.J. Klippenstein, L.B. Harding, Y. Georgievskii, Proc. Combust. Inst. 31 (2007) 221–229.
[71] E. Goos, A. Burcat, B. Ruscic, Ideal Gas Thermochemical Database with updates from Active Thermochemical Tables. <ftp://ftp.technion.ac.il/pub/supported/aetdd/thermodynamics>.
[72] CHEMKIN-PRO 15092, Reaction Design, San Diego, 2009.
[73] K.K. Pant, D. Kunzru, Chem. Eng. J. 67 (1997) 123–129.
[74] J. Kim, S.H. Park, C.H. Lee, B.-H. Chun, J.S. Han, B.H. Jeong, S.H. Kim, Energy Fuels 26 (2012) 5121–5134.
[75] A.C. Davis, N. Tangprasertchai, J.S. Francisco, Chem. Eur. J. 18 (2012) 11296–11305.
[76] P.J. Linstrom, W.G. Mallard, NIST Chemistry Webbook, National Institute of Standard and Technology, Number 69, Gaithersburg, MD, 2005, <http://webbook.nist.gov/>.
[77] J.C. Biordi, C.P. Lazzara, J.F. Papp, Combust. Flame 23 (1974) 73–82.
[78] U. Struckmeier, P. Osswald, T. Kasper, L. Böhling, M. Heusing, M. Köhler, A. Brockhinke, K. Kohse-Höinghaus, Z. Phys. Chem. 223 (2009) 503–537.
[79] P.A. Skovorodko, A.G. Tereshchenko, O.P. Korobeinichev, D.A. Knyazkov, A.G. Shmakov, Combust. Theor. Model. 17 (2013) 1–24.
[80] E.W. Kaiser, W.O. Siegl, D.F. Cotton, R.W. Anderson, Environ. Sci. Technol. 26 (1992) 1581–1586.